

Physicians' Occupational Licensing and the Quantity-Quality Trade-Off*

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Abstract: Occupational licensing is a widespread quality regulation that potentially increases the quality of labor but reduces its quantity. We provide a theoretical framework to empirically quantify this trade-off in public hospitals, where both clinical quality and physician shortages are critical concerns. Using quasi-exogenous variation driven by a recent and unprecedented migration of physicians to Chile, together with growth in domestic medical schools, we show that more physicians improve patient outcomes, while lower-quality physicians—as measured by licensing exam performance—worsen them. Incorporating these findings into our theoretical framework, we evaluate the implications of locally changing the stringency of the licensing policy.

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1 Introduction

The shortage of physicians is a long-standing and increasingly urgent concern. Meeting global healthcare demands is estimated to require at least an additional 7 million physicians, and a deficit of 125 thousand physicians is projected by 2034 for the United States alone ([Haakenstad et al., 2022](#); [Association of American Medical Colleges, 2021](#)).

Licensing requirements in the medical profession have often been identified as a significant contributor to physician shortages, with unclear benefits in terms of quality improvements.¹ Indeed, assessing the quantity-quality trade-off embedded in licensing requirements has proven difficult, as it requires credible estimates of its effects on the quantity and quality of physicians and on the resulting impacts on quality of care. The increasing global migration of physicians has further heightened the impact of licensing policies, potentially altering both the nature and the magnitude of these trade-offs.²

In this paper, we provide a simple framework to highlight the key economic fundamentals that govern the quantity-quality trade-off when deciding the stringency of licensing requirements. From this framework, we derive the empirical objects needed to evaluate the optimality of the licensing requirements and to quantify the net benefits of locally changing them. Using rich administrative data from Chile and a quasi-experimental research design, we estimate those empirical objects and evaluate the optimality of the licensing policy that determines physicians' eligibility to practice.

Chile offers an ideal setting to study the quantity-quality trade-off in physician licensing requirements. First, Chile's licensing exam shares characteristics and objectives that align with examinations used in other countries ([Mena, 2021](#)).³ Second, the number of physicians taking the licensing exam increased fivefold between 2009 and 2018, driven largely by an unprecedented rise in foreign-trained test-takers. In tandem with this surge, passing rates decreased from 92% to 58% during the same period. Third, evaluating the effects of changing the licensing stringency across different phases of the migration wave also allows us to draw more general conclusions about how licensing policies interact with labor market fundamentals, including the quality distribution of applicants,

¹These policies are justified on the grounds that the average patient cannot assess a physician's clinical skill, and the risk to the patient from poor quality is severe ([Johnson, 2026](#)). An influential early critique of licensing in the medical profession appears in [Friedman \(1962\)](#), with a more recent review offered by [Svorny \(2004\)](#). While early discussions centered on healthcare, occupational licensing regulations extend far beyond the medical field: approximately 25% of all US workers are employed in occupations that require an occupational license ([Dillender et al., 2026](#)).

²The number of physicians born and educated outside the US and Canada represented around 20% of the total count of active physicians in the United States in 2021 ([Weiner, 2023](#)). Similarly, in OECD countries, the share of foreign-trained physicians has increased over the past decade, with more than 18% of doctors in 2017/18 having obtained their first medical degree abroad ([Socha-Dietrich and Dumont, 2021](#)).

³Similar exams are the USMLE in the US, the NCE in France, and the NMLE in Japan.

as well as the degree of scarcity.

Our theoretical framework highlights the quantity-quality trade-off embedded in the design of licensing policies. We model hospital care with a production function whose primary inputs are the number and quality of physicians. The planner faces an exogenous distribution of physicians and has a licensing technology that provides a noisy signal of their quality. The planner grants licenses to those with a signal above a threshold. Changing the licensing threshold affects hospital outcomes depending on two sets of elasticities: the output elasticities with respect to the quantity and quality of labor and the elasticity of those inputs with respect to the licensing threshold. The input elasticities, in turn, depend on the fraction of test-takers at the threshold, their score relative to the score of supra-marginal physicians, their probability of matching with the focal hospital, and the precision of the licensing score as a signal of quality (i.e., the signal-to-noise ratio).

Building on our theoretical framework, we estimate the elasticities necessary to gauge how marginal changes in the licensing threshold would affect hospital outcomes in Chilean public hospitals. We first recover output elasticities with respect to both the quantity and the quality of physicians, using mortality as our primary outcome. To address the endogeneity of inputs—pervasive in production-function estimation—we use shift-share instruments (Altonji and Card, 1991; Autor et al., 2013; Borusyak et al., 2022) in a two-stage least squares (2SLS) approach. The instruments for physician quantity and quality leverage the change in both the quantity and average performance of licensing-exam takers, driven by the unprecedented migration of physicians into the country and by the domestic expansion of medical schools.

Our 2SLS estimates show that increases in the number of physicians raise patient volume nearly proportionally, while physician quality has no detectable effect on volume. For mortality, however, both dimensions matter: greater physician availability and higher physician quality are each associated with better patient outcomes. We find that decreasing the number of physicians by 1% increases the in-hospital death rate by 0.67%, while a one standard deviation decrease in average physician quality increases death rates by 17%. Together, these estimates imply that a 10% decrease in the quantity of physicians would increase mortality unless their quality improves by more than 0.39 standard deviations. We further show that both physician quantity and quality matter more for mortality among patients with amenable diseases and that both inputs reduce readmission rates due to complications, an alternative measure of hospital quality.

Next, we proceed by estimating the input elasticities that quantify how both the quantity and quality of physician inputs respond to changes in the licensing threshold. To do so, we specify a dynamic model of score realizations and retake decisions, which maps each physician’s observed

history of scores into a latent quality measure. Our estimates reveal a high signal-to-noise ratio in test scores, implying that scores are informative signals. As a result, adjustments to the licensing stringency can meaningfully change physician quality. We next combine the distribution of test-takers and scores with a matching function between physicians and hospitals. Guided by theory, we approximate this labor market matching as a flexible hospital-specific function of physicians' latent quality and other observables, as well as of observable proxies for labor-market tightness. As such, our estimated labor elasticities account for the fact that the matching function can vary with counterfactual licensing thresholds, since labor market tightness depends on how many physicians pass the exam. Our estimated input elasticities reveal the quantity-quality trade-off for inputs at the margin. In the average hospital-year in our sample, an increase in the threshold that lowers quantity by 10% would increase average quality by 0.37 standard deviations.

Equipped with estimates for the input and output elasticities, we quantify the net benefits of marginally increasing the licensing threshold in the Chilean healthcare context. We begin by assessing the short-run effects of a counterfactual threshold increase 5 years after the onset of the large migration wave. Our results imply that increasing the threshold would have muted effects on patient mortality, as the quantity losses offset the quality gains, while reducing access to care. Our analysis of local changes to the licensing threshold is directly relevant for policymakers who, in practice, consider gradual adjustments close to the current cutoff rather than dramatic reforms. Moreover, our framework naturally extends to guiding a sequence of policy changes: after an initial small adjustment, policymakers can re-evaluate optimality with new data and decide whether to adjust further, maintain the current level, or reverse course.

We also investigate how the short-run effects differ across the different phases of the migration wave. Overall, we find qualitatively similar short-run effects of the policy throughout our sample period, despite the large changes in labor market fundamentals produced by the migration wave. Still, our results help us shed light on how licensing policies interact with labor market conditions. The magnitudes of the input elasticities are non-monotonic over time; the largest impacts would have occurred in the middle of the migration wave when the number of marginal physicians was large and the stock of physicians already hired was still relatively low. Finally, we use our model of scores and retaking behavior to assess the policy's dynamic ("medium-run") effects, taking into account that examinees can retake the exam. Because retaking allows scores to improve over successive attempts, it attenuates the impact of raising the licensing threshold. In fact, more than 80% of the candidates who failed their first attempt in 2013 passed by 2018. As a result, the effect of increasing the threshold on inputs wanes over time, although a permanent increase in the threshold in 2013 still yields measurable impacts in 2018.

Our paper contributes to several strands of the literature. We provide quasi-experimental evidence on the impact of physician quantity and quality on health outcomes. Previous studies have examined the relationship between physician availability and health outcomes, including the link between location-specific mortality and the number of physicians (Finkelstein et al., 2021), as well as the effect of primary care physician supply on infant health (Carrillo and Feres, 2019), and the impact of physician exits on patient mortality (Branco et al., 2026). In this paper, we provide direct evidence that physician quantity influences health outcomes in a tertiary care setting, leveraging quasi-exogenous variation for identification. In addition, we examine physician quality—measured through a licensing exam—and its impact on patient health outcomes. Prior work has documented associations between patient outcomes and various quality proxies: elite medical training (Doyle et al., 2010; Schnell and Currie, 2018), physician value-added measures (Fletcher et al., 2014; Ginja et al., 2025), medical school exit exams (Posso et al., 2026), community-level experiments that replaced mid-level providers with physicians (Okeke, 2023), endowment-funded healthcare modernization (Hollingsworth et al., 2024), and medical school closures (Clay et al., 2025). Also, Fleitas (2020) shows that scores in specialty entry exams correlate with several proxies of quality, including labor market outcomes, treatment costs and length of stay, and patient demand. However, to our knowledge, no study has causally linked licensing-exam performance to patient health outcomes.⁴

We also add to a long-standing debate about the desirability of occupational licensing in healthcare (Friedman and Kuznets, 1945; Friedman, 1962; Svorny, 2004; Kleiner, 2014; Kleiner et al., 2016; Xia, 2021; Dillender et al., 2026) and other settings, particularly in the public sector (Kleiner and Wang, 2026; Angrist and Guryan, 2008; Kleiner, 2011; Larsen et al., 2020; Verboven and Yontcheva, 2024). We make three contributions relative to this literature. First, while most of the occupational licensing literature in healthcare studies the impacts on workers (e.g., wages and employment), we focus on patient outcomes. Notable exceptions include the studies by Anderson et al. (2020) and Noghanibehambari and Fletcher (2023) on the introduction of licensing requirements for midwives in the early twentieth-century United States, Kleiner and Kudrle (2000) who study state-level variation in dentistry licensing, and Oh and Kleiner (2025) who show that the universal license recognition for out-of-state physicians improved access to care.⁵

Second, we provide an empirical framework for understanding and quantifying the quantity-quality trade-off embedded in licensing in the context of the public sector, where wages are rigid and prices

⁴In the context of education, several studies show weak associations between licensing exams and teacher value added (see Clotfelter et al. 2006, 2007, 2010; Goldhaber et al. 2017; Orellana and Winters 2025).

⁵Recent empirical studies outside of healthcare also explore this quantity-quality trade-off. For instance, stricter licensing restricts supply without meaningfully improving quality for certified public accountants (Barrios, 2022) and home improvement professionals (Farronato et al., 2024). For evidence on the trade-off between labor supply and professional discipline in the legal profession, see Chilton et al. (2024) and Rozema (2024).

are fixed.⁶ This framework allows us to evaluate the short- and medium-run effects on relevant outcomes under a counterfactual policy. Close to our work, [Kleiner and Soltas \(2023\)](#) provide a general equilibrium framework to estimate the welfare effect of licensing policies in the US under competitive product and labor markets.⁷ In their framework, labor quality is inferred from wages and equilibrium conditions in the labor market. In contrast, we use mortality as a direct measure of patients’ welfare, and develop a framework to assess the impact of licensing in a setting with rigid wages.

Finally, our work also contributes to recent research on the impact of high-skill migration ([Beerli et al., 2021](#); [Brinatti and Guo, 2023](#); [Dodini et al., 2025](#)) and to the scarce literature estimating production functions and value added in healthcare ([Gaynor et al., 2015](#); [Grieco and McDevitt, 2017](#); [Einav et al., 2025](#)).

The remainder of the paper is organized as follows. Section 2 describes the setting and the data. In Section 3, we provide descriptive evidence to illustrate the main variation leveraged to assess the quantity-quality trade-off. Section 4 presents a stylized model of the licensing problem. In Sections 5 and 6, we present the model and estimates of the output and input elasticities, respectively. We present our counterfactual exercises in Section 7 and conclude in Section 8.

2 Setting and Data

2.1 Institutional Setting

The Chilean Public Healthcare System The healthcare system in Chile is divided into public and private insurers and providers. Our focus is on public providers, which mainly serve patients with public insurance. Public insurance covers approximately 80% of the population and is financed through monthly contributions deducted from labor income, cost-sharing mechanisms, and resources from the general government.⁸

Public health insurance provides coverage within the network of public providers, with varying

⁶The standard framework assumes that restricting labor supply through licensing raises prices for consumers and increases earnings for incumbents ([Johnson, 2026](#)). This view is incomplete in the context of public sector healthcare, where wages and prices are fixed and regulated. Public hospitals account for almost three of every four hospital beds in the average OECD country ([OECD, 2022](#)).

⁷See also [Atal et al. \(2022\)](#) for a general-equilibrium analysis of the welfare effects of minimum quality standards in product markets.

⁸Individuals also have the option to redirect their health contributions toward purchasing private insurance, which covers 15% of the population. The remaining 5% is covered through schemes exclusively for the police and armed forces. In practice, public insurance primarily serves the relatively more disadvantaged population, while wealthier, healthier, and younger individuals tend to opt for private insurance ([Pardo, 2019](#)).

levels of copayment determined by income and family size. Individuals who cannot afford to pay are granted free access to the public system, which ensures nearly universal health coverage in public hospitals. Beneficiaries of the public insurer may also opt for private providers, although the co-payments for such “out-of-network” services are significantly higher than those in the public network.

In our sample, the network of public hospitals comprises 176 hospitals that belong to one of 29 referral regions called Health Services (*Servicios de Salud*).⁹ The referral and counter-referral system also operates at this level. Individuals must register with their local primary healthcare provider, and those requiring specialized care are referred to one of the public hospitals within their region. Referrals follow strict and predetermined guidelines based on diagnosis, location, and other patient demographics.¹⁰

Physician Licensing To work in the healthcare sector, physicians are required to pass the EUNACOM exam (*Examen Único Nacional de Conocimientos de Medicina*).¹¹ The exam was established in 2009 and is aligned with the characteristics and purposes of medical licensing exams in other countries.¹² It comprises a theoretical section and a practical section. The theoretical section consists of a multiple-choice test with 180 questions that cover various areas of medical knowledge. The score reflects the candidate’s absolute performance and is not standardized relative to other test-takers.¹³ To pass, candidates must achieve a score of 51 or higher out of 100 points.

Upon passing the theoretical section, candidates must complete a pass-or-fail practical section, which involves an examination in a real or simulated clinical environment. However, this requirement is waived for candidates with medical degrees from local universities and physicians with a medical degree from a select group of countries with which the Ministry of Health has established

⁹The 176 hospitals correspond to those that are present every year throughout our study period and report deaths at least in one year. We exclude psychiatric hospitals.

¹⁰Patients can also be admitted directly to hospitals through the Emergency Room in cases of emergency.

¹¹Although the exam is not technically mandatory for physicians in private healthcare organizations, it is effectively binding across nearly all healthcare institutions because passing it is a legal requirement in order to treat patients covered by public health insurance, regardless of the treatment location. This explains why, de facto, most private healthcare institutions only consider applications admissible if candidates provide evidence of having passed the EUNACOM. Performance on the EUNACOM is also a determinant for accessing training opportunities, as it plays a key role in admissions to residency programs in the country.

¹²Comparable exams include the Medical Licensing Examination (USMLE) in the US, the Medical Council of Canada Qualifying Examination (MCCQE) in Canada, the National Competency Exam (NCE) in France, the National Medical Licensing Examination (NMLE) in Japan, and the Korean Medical Licensing Examination (KMLE) in Korea. For an in-depth discussion of the exam and evidence of its validity and reliability, see [Mena \(2021\)](#).

¹³A key objective of EUNACOM designers is to make the difficulty of the exam comparable across years. Empirically, we observe that although the scores are not standardized, the distribution of scores at local universities is very similar across years.

bilateral agreements.¹⁴ The practical portion of the exam is largely not binding (Kunakov et al., 2018).

Importantly, passing the EUNACOM automatically validates the medical degrees of foreign-trained physicians and grants them the same job and training opportunities as locally trained physicians.

Physicians’ Wages and Hiring in the Public Sector: Physicians can work in both the private and public sectors. In the private sector, employment operates under standard market dynamics, whereby wages, benefits, and working conditions are negotiated directly between employers and employees. In contrast, in public hospitals, wages are legally regulated and follow a public-sector wage schedule, with annual adjustments based on sector-wide wage revisions.¹⁵

Hiring in public hospitals is decentralized and managed by the regional Health Services within the budgetary constraints established by the Ministry of Health. Health Services determine staffing needs for hospitals, and hospital directors oversee the recruitment process (Muñoz and Otero, 2025). Although passing the EUNACOM exam is sufficient for eligibility, hospitals reportedly consider the candidate’s EUNACOM score an important factor in hiring decisions (EUNAMed, 2021).

2.2 Data

Data Sources Our main analysis combines several data sources: matched employer-employee data for all public hospitals, the national registry of public healthcare providers, data on licensing scores, and individual-level discharge and death records.

Employer-Employee Data for Public Hospitals: We use matched employer-employee administrative records managed by the Ministry of Health, which cover the universe of physicians employed in all public hospitals in Chile between 2011 and 2019 (SIRH, 2019). These data consolidate information from various public health organizations into a unified registry and provide comprehensive records for payroll processing and workforce management, including detailed wage data and employment information. We complement these data with hospital-level characteristics, including size, location, and referral area, among others (DEIS, 2020).

National Registry of Public Healthcare Providers: We use a registry of all physicians (and other healthcare professionals) legally authorized to practice in Chile (RNPI, 2024). The registry is

¹⁴Argentina, Brazil, Colombia, Ecuador, Spain, the United Kingdom, and Uruguay.

¹⁵This is similar to the pay scales for physicians in the NHS, which are determined nationally through negotiations between government bodies and medical unions. In Appendix Figure A.2 we plot the average hourly wages for physicians in public hospitals from 2011 to 2018. Despite the significant growth in the number of physicians, hourly wages remained flat after adjusting for sector-wide remuneration adjustments.

managed by Chile’s Superintendency of Health (*Superintendencia de Salud*) and provides detailed information on registered healthcare professionals, including their nationality and details of their professional degree including title, date of issuance, and the name and country of the granting institution. For degrees obtained abroad, the registry includes the date of revalidation in Chile.

Licensing Exam Scores: We use confidential score records for all physicians who have taken the national licensing exam (EUNACOM). These data were provided directly by ASOFAMECH (*Asociación de Facultades de Medicina de Chile*), the organization responsible for administering and overseeing the exam (ASOFAMECH, 2019). The dataset includes the date and scores of the theoretical portion for the EUNACOM for every attempt made by the universe of physicians who have taken the exam.

Individual-level Discharge and Death Records: We measure outcomes using administrative records of individual-level inpatient events in all public hospitals in Chile from 2011 to 2019 (DEIS, 2019). The data include diagnoses according to the International Classification of Diseases, Tenth Revision (ICD-10 codes), discharge or in-hospital death dates, and patient characteristics such as birth date, gender, residence, and health insurance type. We link these data at the individual level with the universe of death records processed by the Vital Records Office. These records include deaths outside public hospitals but are only available until 2018.

Data Aggregation and Outcomes We construct a hospital-by-year panel dataset using all of the sources of information described above. Our primary measure of care quality is the hospital’s annual mortality rate, defined as the ratio of deaths to admissions in a given year. Following the procedure in Ash et al. (2012), we also construct a risk-adjusted performance measure defined as the ratio between the observed hospital-level mortality rate and the rate predicted based on patients’ risk profiles.¹⁶ To capture deaths occurring outside the hospital, we also consider a 28-day mortality rate, which includes all deaths within 28 days of admission, regardless of place of death. We complement this analysis with additional outcomes, including complication rates and yearly hospital admissions.

Since EUNACOM was introduced in 2009, we impute scores for physicians who did not take the exam.¹⁷ Our imputation procedure assigns scores to physicians from the region of training r working

¹⁶Specifically, we use patient-level data to estimate a logit model of mortality as a function of demographics and diagnoses. The predicted probabilities (patient-level risk scores) are then aggregated to construct hospital-level predicted mortality rates.

¹⁷Before the introduction of EUNACOM, Chile had a voluntary National Medical Examination (EMN) administered in Chilean medical schools from 2003 to 2008. Prior to the EMN, licensing requirements varied: local medical graduates were required to pass the Medical Surgeon Degree Examination, while foreign-trained physicians had to complete a Foreign Medical Qualification Revalidation Examination.

at hospital h by combining the average score of all physicians at hospital h (the grand mean) with the score differential of physicians from region r working at hospital h . To estimate the latter, we run several hospital-level regressions of licensing scores on region-of-origin fixed effects, imposing sum-to-zero constraints on those fixed effects. We then adjust these estimates using empirical Bayes, such that region differentials are shrunk towards zero as the estimates’ precision decreases (Efron and Morris, 1973; Walters, 2024). In Appendix A, we present additional details on our imputation procedure.

Panel A in Appendix Table A.1 summarizes key patient and hospital characteristics. Per hospital-year, there are on average 5,664 inpatient admissions and 78 physicians. The average in-hospital death rate is 3.3%. Panel B describes the evolution of test scores across multiple years, and we discuss them in further detail in the next section.

3 Descriptive Evidence

In line with similar trends in other OECD countries, the number of physicians per capita in Chile has increased, driven by a growing number of domestic graduates and greater reliance on foreign-trained physicians (OECD, 2019). Panel A in Figure I shows that the number of new physicians enrolled in the National Registry of Healthcare Providers rose from around 1,500 in 2011 to almost 3,500 in 2018, with most of the increase explained by an unprecedented inflow of foreign-trained physicians. The share of physicians in Chile who are foreign-trained was 21% in 2017, higher than the OECD average of 16%, but lower than the levels observed in Canada (24%) and the United States (25%).¹⁸ In Appendix B, we document that this increase was unique to physicians and did not extend to other foreign-trained healthcare professionals.

The increasing number of physicians allowed to work in the country is also reflected in the number of physicians taking the licensing exam and their scores. In 2013, relatively few foreign-trained physicians took and passed the exam (i.e., had a score ≥ 51), as shown in Panel B of Figure I. By 2018, the number of foreign-trained physicians taking the licensing exam had increased nearly tenfold, and they outnumbered local graduates by 2.5 to 1, as illustrated in Panel C.

Importantly, not only did the number of foreign-trained physicians increase, but the whole distri-

¹⁸While immigrants represented 2% of the population in 2011, this figure rose to 8% by 2022 (INE, 2023), largely driven by the mass migration of Venezuelans to Chile and other countries in the region, which began in late 2015 following the collapse of the Venezuelan economy. Venezuelan migration to Chile and other countries in the region has been leveraged as a shock to explore the impacts of migration on various outcomes, including crime (Ajzenman et al., 2023); labor markets (Lebow, 2022; Olivieri et al., 2022; Bahar et al., 2024); and discrimination (Groeger et al., 2024), among others.

bution of scores shifted to the left, and by 2018 the distribution of foreign-trained physicians was centered around the licensing threshold. This indicates that while many successfully validated their medical degree, many were unable to do so because the licensing threshold was a binding constraint for them. Panel B of Appendix Table A.1 shows that the proportion of all test-takers who achieved a passing score declined from 92% in 2009 to 58% in 2018. Meanwhile, the number of individuals scoring within the [51–56) range increased substantially, from 70 in 2009 to 856 in 2018, which reflects the growing mass of physicians who qualify with a score very close to the threshold.

Figure II shows aggregate trends in public hospitals’ inputs and outcomes, and previews some of the key results of our empirical analysis. There was a large increase in the number of physicians working in public hospitals in Chile during our period of analysis. Panel A shows that by 2018, the number of full-time physicians in public hospitals had risen by about 50% relative to 2011, reflecting the influx of newly registered physicians.

Concurrently with the increase in the quantity of physicians, the average scores of physicians working in public hospitals decreased during the period. Panel B of Figure II shows the distribution of the average hospital scores across all public hospitals in the country. We observe that the whole distribution significantly shifted to the left. On average, the scores of physicians in public hospitals decreased by 0.5 standard deviations between 2013 and 2018.

The aggregate trends in mortality provide suggestive evidence for a quantity-quality trade-off at play. Panel C of Figure II shows the trend in aggregate death rates across public hospitals over time as the number of physicians increased but their scores decreased. At face value, there is no visible change in average mortality over time. Two alternative explanations can rationalize these findings. Either the quantity and quality of physicians do not affect mortality, or the effect of increasing the number of physicians was offset by the decrease in their average quality. Understanding the magnitude of these effects and the potential trade-off embedded in physician licensing is what we study in detail in the remainder of the paper.

4 A Simple Licensing Problem

We begin with a simple theoretical framework to study the economic fundamentals that govern the effects of licensing in our context. We focus on a particular margin: how the stringency of the requirement, as determined by the passing score of the licensing exam, changes quantity and quality of labor and ultimately patients’ outcomes.

Consider a public hospital endowed with a production function F . The planner faces a distribution

of physicians with latent quality θ and has access to a licensing technology based on a noisy signal of quality, s . The planner grants licenses to those with $s > \underline{s}$, where $\underline{s}$ is the policy variable determining the licensing threshold (passing score). The output of interest is an increasing function of labor $L(\underline{s})$ and a quality index $Q(\underline{s})$,

$$Y(\underline{s}) = F(L(\underline{s}), Q(\underline{s})),$$

where $F_L > 0$ and $F_Q > 0$. The outcome Y can represent population survival (the opposite of mortality), which increases with L through, for example, shorter waiting times and less deterioration of health capital and also increases with Q through better quality of care.

This production function generates a quantity-quality trade-off in the licensing policy, as long as there is positive mass around $\underline{s}$ and the quality input is positively related to the cutoff: $Q(\underline{s})' > 0$. Specifically, the elasticity of the outcome with respect to the licensing threshold, $\eta_{\underline{s}}^Y \equiv \frac{\partial Y}{\partial \underline{s}} \frac{\underline{s}}{Y}$, is equal to

$$\eta_{\underline{s}}^Y = \underbrace{\eta_L^Y \eta_{\underline{s}}^L}_{\text{Licensing Quantity Effect}} + \underbrace{\eta_Q^Y \eta_{\underline{s}}^Q}_{\text{Licensing Quality Effect}}, \quad (1)$$

where η_L^Y and η_Q^Y are the output elasticities with respect to the quantity and quality inputs, respectively, and $\eta_{\underline{s}}^L$ and $\eta_{\underline{s}}^Q$ denote the elasticities of each input with respect to the licensing threshold. The impact of the policy variable $\underline{s}$ on the outcome $Y(\underline{s})$ is fully characterized by these four elasticities. On the one hand, increasing the threshold decreases output by reducing the quantity of labor. This *Licensing Quantity Effect* depends on the output elasticity with respect to quantity, and the elasticity of labor with respect to the threshold. On the other hand, increasing the threshold increases output by improving quality. This *Licensing Quality Effect* depends on the output elasticity with respect to quality and the elasticity of quality with respect to the threshold.

The ratios of input and output elasticities are sufficient to determine the sign of the effect of changing the licensing threshold on the outcome as

$$\eta_{\underline{s}}^Y > 0 \iff \eta_L^Y / \eta_Q^Y < -\eta_{\underline{s}}^Q / \eta_{\underline{s}}^L. \quad (2)$$

Equation (2) simply states that it is beneficial to increase the licensing threshold if and only if the ratio of output elasticities between quantity and quality is smaller than the ratio of the associated changes in those inputs. In other words, it is more beneficial to increase the threshold the higher the return to quality relative to the return to quantity, and the higher the gains in quality relative

to the losses in quantity.

Parameterizing the model allows us to gain further insights into the microfoundations for the elasticities that govern the quantity-quality trade-off in licensing. Let the signal s have a distribution with density $h(s)$ and total mass m , and further assume that $s = \theta + \epsilon$, with $\epsilon \sim N(0, \sigma_\epsilon^2)$ and $\theta \sim N(\mu_\theta, \sigma_\theta^2)$; defining the signal-to-noise ratio as $\text{SNR} \equiv \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\epsilon^2}$. Physicians who pass the exam can either work at the hospital or go to an outside option. We let $p(s|\underline{s})$ denote the share of physicians with score s who match with the hospital; this may depend on the licensing threshold $\underline{s}$ through equilibrium effects in the labor market. The total labor function is therefore $L(\underline{s}) = m \int_{\underline{s}}^{\infty} h(s)p(s|\underline{s})ds$. Let the quality index be equal to average quality, $Q(\underline{s}) = \bar{\theta}(\underline{s})$, with $\bar{\theta}(\underline{s}) = \frac{m}{L(\underline{s})} \int_{\underline{s}}^{\infty} \hat{\theta}(s)h(s)p(s|\underline{s})ds$, where $\hat{\theta}(s) \equiv E[\theta|s]$. Finally, the production function has the form $F(L, \bar{\theta}) = \exp(\alpha_{\bar{\theta}}\bar{\theta}) \cdot L^{\alpha_L}$, where the term $\exp(\alpha_{\bar{\theta}}\bar{\theta})$ represents a quality-adjusted productivity, and α_L represents the returns to the scale of labor.

Under this parameterization, the elasticity of output with respect to the licensing threshold $\underline{s}$ can be written as

$$\eta_{\underline{s}}^Y = \alpha_L \cdot \eta_{\underline{s}}^L + \alpha_{\bar{\theta}} \cdot \tilde{\eta}_{\underline{s}}^{\bar{\theta}},$$

where $\eta_{\underline{s}}^L$ is the elasticity of labor with respect to the licensing threshold and $\tilde{\eta}_{\underline{s}}^{\bar{\theta}}$ is the semi-elasticity of quality with respect to the threshold. Moreover, in the absence of equilibrium effects in the labor market—i.e., when $\partial p(s|\underline{s})/\partial \underline{s} = 0$ —the elasticities can be expressed as simple functions of the underlying model parameters.¹⁹ The elasticity of labor with respect to the threshold is

$$\eta_{\underline{s}}^L = \frac{-m \cdot h(\underline{s}) \cdot p(\underline{s}|\underline{s}) \cdot \underline{s}}{L},$$

which depends on the mass of workers at the threshold who would match with the hospital, as a fraction of workers above the threshold. In turn, the semi-elasticity of quality with respect to the licensing threshold is

$$\tilde{\eta}_{\underline{s}}^{\bar{\theta}} = -\eta_{\underline{s}}^L \cdot \mathbb{E}[\theta(s) - \theta(\underline{s})|s > \underline{s}],$$

which can be written as

$$\tilde{\eta}_{\underline{s}}^{\bar{\theta}} = -\eta_{\underline{s}}^L \cdot \text{SNR} \cdot (\mathbb{E}[s|s > \underline{s}] - \underline{s}).$$

In sum, the semi-elasticity of quality depends on the share of marginal workers, $-\eta_{\underline{s}}^L$; the effect of

¹⁹In our empirical model, we allow for equilibrium effects to affect the matching probabilities.

marginal workers on average *scores*, $(\mathbb{E}[s|s > \underline{s}] - \underline{s})$; and the precision of the signal, SNR.

On net, the elasticity of outcome with respect to the threshold is therefore

$$\eta_{\underline{s}}^Y = \eta_{\underline{s}}^L \cdot (\alpha_L - \alpha_{\bar{\theta}} \cdot \text{SNR} \cdot (\mathbb{E}[s|s > \underline{s}] - \underline{s})),$$

where the first term is the share of marginal workers, and the second term is the net effect of a marginal worker on production. The quantity effect of an extra worker is α_L . However, the marginal worker decreases average quality by $\text{SNR} \cdot (\mathbb{E}[s|s > \underline{s}] - \underline{s})$. This term, multiplied by the return on quality, gives the quality effect of the marginal worker.

Under this parameterization, it is beneficial to increase the licensing threshold as long as

$$\frac{\alpha_L}{\alpha_{\bar{\theta}}} < \text{SNR} \cdot (\mathbb{E}[s|s > \underline{s}] - \underline{s}) \equiv R(\underline{s}), \quad (3)$$

where $R(\underline{s}) = -\tilde{\eta}_{\underline{s}}^{\bar{\theta}}/\eta_{\underline{s}}^L$ is the marginal rate of transformation between quantity and quality, and thus represents the trade-off embedded in the licensing problem: by decreasing the quantity of labor by 1%, the planner can increase its quality by $0.01 \cdot R(\underline{s})$. Under this parameterization, this increase in quality is the product of the distance between the average score and the marginal score and the precision of the signal. Increasing the licensing threshold increases output if and only if the ratio between the output elasticity with respect to quantity and the output elasticity with respect to quality is lower than $R(\underline{s})$.

Figure III provides a graphical depiction of the licensing problem and the net benefits of increasing the licensing threshold. Different licensing thresholds yield different combinations of available quantity and quality, and trace a decreasing relationship between the two as depicted by the gray curve. The slope of this relationship at a given licensing threshold $\underline{s}$ corresponds to $R(\underline{s})^{-1}$. The production function is depicted by the solid black line, with a slope that reflects the marginal rate of technical substitution between quantity and quality, $\alpha_{\bar{\theta}}/\alpha_L$. The net effect of changing the licensing threshold at the margin depends on the relative magnitude of the slopes at the current threshold. In the example depicted in Figure III, the increase in quality achieved by a 1% decrease in quantity is higher than the increase in quality needed to leave output unchanged. In this case, increasing the licensing threshold would improve outcomes. A key goal of the empirical analysis is thus to quantify these two slopes, which are the empirical objects needed to identify the effects of

changing the threshold on outcomes.²⁰

5 Output Elasticities

In this section, we introduce the hospital production function for healthcare, present our instrumental variables approach to estimate the output elasticities, and discuss our estimation results.

We focus on mortality as the primary outcome of interest. Recent work shows that physician quality plays a central role in shaping mortality outcomes (Okeke, 2023; Ginja et al., 2025), and that improvements in physician quality can, in some contexts, partially offset reductions in physician supply (Clay et al., 2025). Multiple channels may underlie the relationship between physicians’ quantity and quality and patient mortality. On the one hand, having a larger pool of doctors may generate economies of scale that allow for more intensive monitoring and faster intervention when complications arise, thereby improving outcomes and lowering death rates (Hoot and Aronsky, 2008; Hoe, 2022). A greater presence of physicians might also reduce treatment delays for inpatients—a factor that is especially crucial in life-threatening circumstances.²¹ On the other hand, licensing policies are justified on the grounds that the performance of physicians on licensing exams is a good predictor of patients’ outcomes. Clinically stronger doctors may diagnose and manage complex conditions more accurately, make timely decisions, and consistently follow evidence-based practices. Such expertise may also lower the risk of complications, curtail treatment errors and misdiagnoses, and ultimately improve patient survival.

We therefore hypothesize that licensing requirements create a quantity–quality trade-off with direct consequences for mortality. To measure this trade-off empirically, we model patient mortality as a function of both the number of practicing physicians and their quality as inferred from the licensing exam.

²⁰The optimal licensing threshold $\underline{s}^*$ can be implicitly characterized with a simple expression:

$$\underline{s}^* = \mathbb{E}[s | s > \underline{s}^*] - \frac{\alpha_L}{\alpha_{\bar{\theta}}} \cdot \frac{1}{\text{SNR}}.$$

The optimal threshold increases with the mean score. A higher mean allows the planner to obtain more quantity for any given level of quality. Also, the optimal threshold decreases with the output elasticity with respect to labor and increases with the output semielasticity with respect to quality. It is also lower when the licensing technology is imprecise (i.e., low signal-to-noise ratio), as in such cases a higher threshold can only weakly differentiate between high- and low-quality physicians.

²¹Earlier studies show that quicker access to healthcare could influence mortality (see, e.g., Prentice and Pizer, 2007; Pizer and Prentice, 2011; Nikolova et al., 2016; Gruber et al., 2023).

5.1 Health Outcomes Model

In period t , each hospital j produces an outcome y_{jt} with the production function

$$y_{jt} = \exp(\mathbf{O}'_{jt}\boldsymbol{\beta}_O) \times \exp(\alpha_\theta Q(\theta_{jt})) L_{jt}^{\alpha_L} \times \exp(\nu_{jt}), \quad (4)$$

where L_{jt} is the quantity of physicians, and $\mathbf{O}_{jt}$ includes other determinants of hospital output such as other labor, capital, patient case-mix, and other unobserved drivers of productivity. There is also a productivity term, $\exp(\alpha_\theta Q(\theta_{jt}))$, which depends on an index of physician quality, $Q(\theta_{jt})$. This quality index captures how hospital outcomes are affected by the distribution of the quality of physicians at hospital j in period t . Finally, ν_{jt} is a random shock. The parameters of interest for the quantity-quality trade-off are α_L and α_θ , which correspond to the output-specific elasticities with respect to the quantity and quality of physicians, respectively. In Appendix C, we provide a microfoundation for this production function by deriving it from a patient-level model in which physician quantity and quality affect outcomes both directly and indirectly through waiting times.

Empirical implementation of this model requires a stance on the functional form for the quality index. In our main specification, we assume that $Q(\theta_{jt}) = \bar{\theta}_{jt}$; i.e., the quality index is the mean of the quality distribution in hospital j in period t . We also specify $\mathbf{O}'_{jt}\boldsymbol{\beta}_O$ to be a linear combination of hospital fixed effects ρ_j , year fixed effects that vary with hospital type, $\gamma_{f(j)t}$, the log number of beds, B_{jt} , demographics of each hospital's patients D_{jt} , and an unobservable ω_{jt} .²²

$$\mathbf{O}'_{jt}\boldsymbol{\beta}_O = \rho_j + \beta_D D_{jt} + \beta_B B_{jt} + \gamma_{f(j)t} + \omega_{jt}.$$

Let $\bar{\epsilon}_{jt} \equiv \sum_{i \in I_{jt}} \epsilon_{it}/L_{jt}$ and $\bar{s}_{jt} \equiv \sum_{i \in I_{jt}} s_{it}/L_{jt}$ denote the average measurement error and score in hospital j in period t , respectively. Substituting $\theta_i = s_i - \epsilon_i$ into Equation (4) and taking the natural logarithm yields

$$\ln(y_{jt}) = \alpha_L \ln(L_{jt}) + \alpha_\theta \bar{s}_{jt} + \gamma_{f(j)t} + \rho_j + \beta_D D_{jt} + \beta_B B_{jt} + u_{jt}. \quad (5)$$

This is our estimating equation to recover α_L and α_θ —the output-specific elasticities with respect to the quantity and quality of physicians, respectively. The error term, u_{jt} , consists of three components: an unobserved productivity shock potentially known before input choices ω_{jt} ; measurement

²²In our empirical implementation, we follow the regulator's categorization of hospitals into three types—low, medium, and high complexity—depending on their size and the range of medical services they offer. The vector D_{jt} includes the shares of female and foreign patients, as well as the patient shares across eight age bands (0–29, then 10-year increments to 90+), and insurance types grouped by copayment levels.

error in quality, $-\alpha_\theta \bar{\epsilon}_{jt}$; and an unobserved productivity shock that occurs after input decisions, ν_{jt} . To address the identification challenges posed by unobserved productivity shocks and the measurement error in latent quality, we implement an instrumental-variable approach (Agostinelli and Wiswall, 2025).

5.2 Instruments and Identification

In this subsection, we define our shift-share instruments and present reduced-form evidence to illustrate the identifying variation leveraged by our approach.

We begin by constructing a Bartik-style instrument for physician quantity, Z_{jt}^L . This instrument leverages the variation in the number of licensing-exam takers who clear the cutoff of the licensing exam, driven by the unprecedented migration of physicians into the country and by the domestic expansion of medical schools. Concretely, we calculate the percentage change in the number of test-takers who clear the cutoff from each region of training c , M^c . Formally, and following the notation in Acemoglu and Restrepo (2020), we define $\% \Delta \text{Test-takers}_{(t_0, t_1)}^c \equiv (M_{t_1}^c - M_{t_0}^c) / M_{t_0}^c$, where $(t_0, t_1) = (2011, 2015)$ if $t < 2015$ and $(t_0, t_1) = (2015, 2019)$ if $t \geq 2015$. In order to lend predictive power to the instrument, we interact the shift by region of training with the share of physicians from region c who worked at hospital j the year before, defined as $\text{Share Physicians}_{cjt-1} \equiv \frac{L_{cjt-1}}{\sum_c L_{jct-1}}$.²³ The instrument for physician quantity is thus:

$$Z_{jt}^L = \sum_c \% \Delta \text{Test-takers}_{(t_0, t_1)}^c \times \text{Share Physicians}_{cjt-1}.$$

This instrument captures differences in hospitals' exposure to changes in the supply of physicians trained in each region. To illustrate how this variation affects hospitals, we examine the reduced-form responses of physician quantity, physician quality, and in-hospital mortality to Z_{jt}^L . Specifically, we estimate the following version of Equation (5):

$$y_{jt} = \pi^y Z_{jt}^L + \gamma_{f(j)t} + \rho_j + \beta_D D_{jt} + \beta_B B_{jt} + \beta_P P_{jt} + u_{jt},$$

where y_{jt} denotes the log number of physicians, their average licensing exam score, or the log in-hospital death rate. We standardize Z_{jt}^L so that π^y measures the change in each outcome associated

²³We use the lagged share to maximize the predictive power of the instrument (Borusyak et al., 2025). Given that before the migration wave, very few foreign doctors were licensed in Chile and they were concentrated in a few nationalities, using the pre-migration share would result in shares close to zero and consequently, very weak instruments.

with a one-standard-deviation increase in the instrument. Figure IV presents the corresponding binned scatterplots, together with the estimated coefficients $\hat{\pi}$ and their standard errors. Panel A shows that hospitals more exposed to increases in physician supply experienced a significant increase in the number of physicians, while Panel B shows a decline in their average licensing exam scores. Panel C reports an estimated effect on mortality close to zero.

The reduced form coefficients imply that the shock changes the two inputs in opposite directions at a ratio of about $\pi^{\bar{s}}/\pi^L = -0.077/0.024 \approx -3.2$. Under Equation (5), the reduced-form effect on mortality satisfies $\pi^{\text{mort}} = \alpha_L \pi^L + \alpha_{\theta} \pi^{\bar{s}}$. Our near-zero estimate therefore suggests an approximate restriction on the relative effects of physician quantity and quality: If $\alpha_{\theta} \neq 0$, then $-\pi^{\bar{s}}/\pi^L = \alpha_L/\alpha_{\theta}$. Together with the policy-induced quantity–quality trade-off, $R(\underline{s})$, this ratio determines the direction of the mortality effect of increasing the licensing threshold, as shown by Equation (3).

This single-instrument exercise, however, does not separately identify α_L and α_{θ} : the near-zero mortality response could reflect, either offsetting quantity and quality effects or small effects of both inputs. To distinguish between these possibilities, we introduce a second instrument that shifts average physician scores. Together, the two instruments allow us to separately identify the effects of physician quantity and quality. This distinction is necessary for evaluating counterfactual policies that might alter these two inputs in proportions different from the observed labor-supply shock.

The instrument for physicians’ quality uses the same share component but replaces the shift component with the change in the average scores of test-takers who clear the cutoff from each region of training c , S^c . Formally, we define $\Delta \text{Avg. Quality}_{(t_0, t_1)}^c \equiv \bar{S}_{t_1}^c - \bar{S}_{t_0}^c$, where $(t_0, t_1) = (2011, 2015)$ if $t < 2015$ and $(t_0, t_1) = (2015, 2019)$ if $t \geq 2015$. The instrument for physician quality is thus:

$$Z_{jt}^{\bar{s}} = \sum_c \Delta \text{Avg. Quality}_{(t_0, t_1)}^c \times \text{Share Physicians}_{cjt-1}.$$

With two endogenous variables, identification of the effects of physician quantity and quality requires the instruments to generate sufficiently distinct variation in the two inputs. To provide a visual representation of this identifying variation, we use the first-stage and reduced-form relationships to map each hospital into the combination of physician quantity and quality predicted by the instruments and examine how mortality varies across these predicted input bundles.

Consider the predictions of the first-stage regressions embedded in the model of Equation (5). Specifically, for each input $I \in \{\ln(L), \bar{s}\}$ we use the first-stage coefficients $\hat{\gamma}_s^I$ and $\hat{\gamma}_L^I$ to predict $\hat{I}_{jt} = \hat{\gamma}_s^I Z_{jt}^{\bar{s}} + \hat{\gamma}_L^I Z_{jt}^L$. These predictions isolate the exogenous variation in the inputs generated

by the instruments. Similarly, to characterize the variation in mortality generated by the instruments, we estimate the reduced-form version of Equation (5); that is, we replace the inputs with the instruments as independent variables and use the resulting coefficients to predict mortality: $\widehat{\ln(\text{mortality rate}_{it})} = \hat{\gamma}_S Z_{jt}^S + \hat{\gamma}_L Z_{jt}^L$.

The instruments generate substantial variation across hospitals in predicted physician quantity and quality. Figure V plots each hospital’s average (over time) predicted quantity and quality, with marker sizes proportional to hospital size at the beginning of our sample. We color hospitals according to whether their mortality rate, as predicted by the instruments, falls in the top, middle, or bottom tercile.

The gray dots correspond to hospitals in the middle tercile of predicted mortality. Because predicted mortality varies relatively little within this group, the fitted relationship between predicted quality and quantity provides a visual approximation to an iso-mortality locus. Consistent with the preceding reduced-form results, the fitted slope is approximately -3.8. The figure also illustrates how predicted mortality varies across predicted input bundles. Hospitals lying below the iso-mortality locus—those for which the predicted decrease in quality is large relative to the predicted increase in quantity—have higher predicted mortality. Conversely, hospitals lying above the iso-mortality locus have lower predicted mortality.

The preceding exercises illustrate how the instruments shift physician quantity and quality and how the resulting variation relates to hospital mortality. A causal interpretation of these relationships, however, requires the shift component of the instruments to be as good as randomly assigned with respect to unobserved determinants of hospital outcomes (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022, 2025). In our setting, this assumption pertains to the shift-level variation generated by the migration shock and the expansion of local medical schools. Supporting this assumption, Appendix Figure A.8 shows that neither the quantity nor the quality shocks predict predetermined hospital characteristics.²⁴

In the next subsection, we use these instruments as sources of exogenous variation to estimate the output elasticities of physician quantity and quality.

²⁴We further assess the robustness of our estimates in the next subsection, including by recentering the instruments to account for potentially non-random exposure to these shocks (Borusyak and Hull, 2023).

5.3 Estimation and Results

Two-Stage Least Squares Results Table I shows the results from our main 2SLS model, the empirical analog of Equation (5).²⁵ We present point estimates and exposure-robust standard errors clustered by physicians’ region-of-training. To obtain these standard errors—which are shown to be asymptotically valid (Adão et al., 2019)—we estimate the transformed shock-level regression proposed by Borusyak et al. (2022). All regressions are estimated at the hospital level over the 2012-2019 period and are weighted by the number of inpatients in 2011.

Column (1) reports results for the log number of inpatient admissions. We find an elasticity with respect to the number of physicians of 1.01, suggesting that the number of patients admitted increases almost proportionally with the number of physicians. We also document no impact of physician quality on the number of admissions.

Column (2) shows the results on our main outcome of hospital quality: in-hospital death rates. We find that (i) increasing the number of physicians by 1% decreases the in-hospital death rate by 0.67% and (ii) a one standard deviation increase in average physicians’ quality decreases death rates by 17%. The implied ratio of the output elasticities is 3.9, which is very close to the ratio from reduced form estimates and the slope of the iso-mortality locus discussed in the previous subsection. In Column (3), we use the 28-day mortality rate as the outcome and find very similar effects to those reported for in-hospital deaths only.²⁶

One potential explanation for the previous findings is that the marginal patients—those who were previously not admitted—may be healthier, so a shift in patient composition could be driving the results (Kelly and Stoye, 2020). To explore this hypothesis, Column (4) examines the impact on the risk-adjusted mortality rate, which, following the UK’s NHS (NHS Digital, 2016), is defined as the ratio between the actual hospital-level death rate and the predicted death rate—i.e., a ratio above 1 indicates more deaths than predicted, and a ratio below 1 indicates fewer deaths than predicted.²⁷ Reassuringly, the effects of both physician quantity and quality on this risk-adjusted measure are very close to those in Columns (2) and (3).

²⁵We assume constant treatment effects, following standard practice in the SSIV literature (Borusyak et al., 2022, 2025). This assumption also keeps identification transparent, as recent work shows that with multiple treatments, linear regressions and multivariate 2SLS can deliver contaminated and hard-to-interpret heterogeneous treatment effects, unless strong additional assumptions are imposed (Blandhol et al., 2026; Bhuller and Sigstad, 2024; Goldsmith-Pinkham et al., 2024). For comparison, Appendix Table A.4 presents the corresponding OLS estimates.

²⁶Because we observe death records only through 2018, our 28-day mortality measure includes both in-hospital and out-of-hospital deaths for inpatients admitted between 2012 and 2018, but only in-hospital deaths for those admitted in 2019.

²⁷We calculate the expected deaths by fitting a logit model for death outcomes at the inpatient level, controlling for patient demographics and diagnosis group, as per the enhanced Elixhauser comorbidity index (Elixhauser et al., 1998; Quan et al., 2005).

To investigate the mechanisms driving our mortality results, we categorize in-hospital deaths based on the concept of amenable mortality—deaths from conditions that are broadly recognized as avoidable in the presence of timely and effective medical care. We restrict the discharge sample to ICD-10 three-digit codes corresponding to acute care conditions amenable to physician treatment and aggregate the data to the hospital-by-year level.²⁸ The estimates in Column (5) indicate that both physician quantity and quality substantially reduce mortality for this amenable subset. Importantly, these effects are larger in magnitude than those estimated for overall in-hospital mortality in Column (2). Conversely, Column (6) reports modest and statistically insignificant effects of both physician quantity and quality among patients with non-amenable conditions.

Finally, Column (7) focuses on the readmission rate due to complications as an alternative measure of hospital quality. We define this rate as the number of patients discharged from the hospital who are later (within 3 months) readmitted for inpatient care, at any hospital, due to an ICD-10 code related to infections, hemorrhage, or other complications, divided by the total number of admissions. Consistent with the impacts on mortality, we find that physician quantity and quality help to reduce the complication rate.

Robustness Checks We conduct several exercises to assess the robustness of our output-elasticity estimates. First, we implement the recentering approach by [Borusyak and Hull \(2023\)](#) to address the possibility that hospitals are non-randomly exposed to the shocks embedded in our shift-share instruments.²⁹ Appendix Table A.5 reports estimates using these recentered instruments. The results are very similar to our baseline findings: physician quantity continues to increase inpatient admissions and reduce mortality, while physician quality continues to have little effect on admissions but substantially lowers mortality.

We also consider two alternative measures of physician quality. First, we re-estimate our baseline specification using the average score among exam takers only, rather than the staff-wide average that includes imputed scores for older physicians who did not take the exam. Appendix Table

²⁸Our classification adapts the concept of healthcare-amenable mortality ([Nolte and McKee, 2004](#)) to our setting. We define amenable conditions ex ante as acute, time-sensitive inpatient diagnoses for which short-run survival is plausibly responsive to clinical intervention. These include acute cardiovascular and cerebrovascular events (I21, I22, I26, I27, I46, I50, I60–I64); severe respiratory infections and respiratory failure (J13, J15, J18, J22, J80, J96); sepsis and other severe bacterial infections (A40–A41, A49); acute kidney failure (N17, N19); shock, systemic inflammatory response, and acute cardiopulmonary distress (R57, R65, R09); and major abdominal and trauma emergencies (K25–K27, K35, K56–K57, K65, K85, S06, S26, S36, S72). Conditions not classified as amenable correspond to the remaining ICD-10 codes in the patient-level data.

²⁹We compute the expected instruments under the assignment mechanism analytically: for each hospital and year, the expected quantity and quality instruments are the shift-share aggregates of the mean shocks within year-by-foreign-trained status cells, which we subtract from the realized instruments. This procedure removes the component of the instruments that is predictable from hospitals’ non-random exposure to the shock assignment process.

A.6 reports the resulting estimates. Although the estimated importance of quality for treatment outcomes is attenuated,³⁰ the qualitative patterns are in line with our main specification. Second, we consider an alternative measure of quality $Q(\theta_{jt})$ defined as the share of physicians in hospital j and period t who have scores below the sample median. As shown by Appendix Table A.7, results remain qualitatively unchanged: a higher share of lower quality physicians is associated with higher mortality.

Finally, an additional potential concern with our empirical strategy is that our instruments may also capture changes in other healthcare inputs—for instance, if the migration wave increased the number of other foreign healthcare workers beyond physicians. However, as discussed earlier and shown in Appendix B, the migration shock did not alter the share of other foreign healthcare workers, based on data from the Registry of Healthcare Providers and public hospitals’ employer-employee records. In addition, Appendix D shows that the ratio between physician and other healthcare workers remained remarkably stable over time.

6 Input Elasticities

As shown by the input-elasticity expressions in Section 4, the effects of changing the licensing threshold depend on three empirical objects. First, the mass of test-takers at the threshold determines how many fewer physicians are licensed. Second, the matching of physicians to hospitals determines how this loss of physicians is borne across hospitals and, therefore, hospital-specific quantity elasticities η_s^L . Finally, the mapping between test scores and physician quality translates these employment changes into quality changes, resulting in hospital-specific quality elasticities $\tilde{\eta}_s^{\bar{\theta}}$.

We recover these objects with a dynamic model of scores and a labor-market matching model.

6.1 Scores and Labor Market Matching Model

Modeling the dynamics of scores serves two purposes. First, it allows us to infer physicians’ latent quality from their score histories. Second, it allows us to quantify changes in the mass of test-takers under counterfactual scenarios after accounting for the possibility of retaking.

³⁰This is expected: observed scores are more dispersed than imputed ones (standard deviation 7.0 versus 6.1), and our instrument-constructed from changes in exam takers’ scores-is mechanically more tightly linked to the exam-based measure than to staff-wide quality. The exam-based specification can therefore be interpreted as a conservative lower bound on the effect of overall physician quality.

Score Evolution Each physician i is characterized by her latent quality θ_i . Her observed score history is determined by score realizations in each attempt, as well as retaking decisions after failing. On average, scores improve over attempts, and thus we model the score of the ℓ^{th} attempt as a noisy measure of latent quality and test-taking ability, $\Gamma_{i\ell}$, which can change over attempts as in Gilraine and Penney (2025).³¹ We let latent quality follow a two-component mixture distribution whose parameters depend on the physician’s region of training (or “type”) $\tau_i \in \{\text{National, Foreign}\}$ and entry cohort t_i^0 , as successive cohorts of foreign-trained physicians arriving during the migration wave may be drawn from different distributions of latent quality.³² We also let score precision vary by type and entry cohort. This specification aims to accommodate the fact that Nationals are often trained in systems that prepare them for the exam, while foreign-trained physicians typically take it years after graduating. Precision may also change across cohorts as the composition and exam preparation may shift over time due to changes in the composition of applicants.³³ Finally, as average score gains over attempts are decreasing and convex (see Figure A.5, Panel A), we assume that test-taking ability—which we normalize to zero in the first attempt—improves with exponential decay.

Formally, the model of realized scores in the ℓ^{th} attempt for individual i is given by

$$\begin{aligned} s_{i\ell} &= \theta_i + \Gamma_{i\ell} + \epsilon_{i\ell}, \\ \theta_i \mid \tau_i, t_i^0 &\sim \text{MixN}_2(\tau_i, t_i^0), \\ \epsilon_{i\ell} \mid \tau_i, t_i^0 &\sim N(0, \sigma_{\epsilon, \tau_i, t_i^0}^2), \\ \Gamma_{i1} &= 0, \quad \Gamma_{i\ell} = \sum_{k=2}^{\ell} \gamma e^{-\rho(k-2)}, \quad \ell \geq 2. \end{aligned}$$

The term t_i^0 is i ’s entry cohort, $\sigma_{\epsilon, \tau_i, t_i^0}^2$ is the type- and cohort-specific variance of the noise, γ captures the average improvement in test-taking ability between the first and second attempt, and ρ governs the average rate at which improvements in test-taking ability decrease over subsequent attempts.³⁴ Modeling the score process allows us to account for selection in estimating the model parameters.³⁵

³¹We assume that quality is constant over time, which implies that physicians cannot experience quality gains over attempts. To support this assumption, in Appendix E we use regression discontinuity designs to show that physicians who retake the exam around the passing threshold do not experience gains on quality proxies.

³²Appendix G provides the full parameterization of this mixture and the corresponding posterior-quality formula.

³³Both the composition of origin countries for foreigners and university of training for nationals changed over our study period as documented in Section 3.

³⁴In principle, γ and ρ could be type-specific. However, since most nationals pass the test on the first attempt, we lack the power to identify parameters for nationals and foreigners separately. This limitation is largely inconsequential for our counterfactual analysis, as nationals represent a small fraction of candidates who fail.

³⁵As discussed by Gilraine and Penney (2025), selection occurs because $\epsilon_{i\ell}$ is not mean zero among individuals

Retaking The second component of the model of scores is the retaking behavior. Modeling retaking behavior allows us to construct the dynamic moments used for the estimation of γ and ρ as discussed in Section 6.2. It also allows us to incorporate potential changes in the mass of test-takers with counterfactual thresholds and study the medium-run consequences of changing the licensing stringency.³⁶

As shown in Figure A.5, Panel B, the retaking probability decreases with the gap between the last score and the threshold. The model of score determination provides a simple rationale: a worse past performance is indicative of a lower quality, and thus of lower passing probabilities in subsequent attempts.

Formally, physicians who fail the exam on attempt ℓ retake it with probability

$$P(\text{retake}|\bar{s}_{i\ell}, \ell, \tau_i, e_i) = \frac{e^{\beta_{0,\tau_i} + \beta_{\text{att},\tau_i}\ell + \beta_{d,\tau_i}(\underline{s} - \bar{s}_{i\ell}) + e_i}}{1 + e^{\beta_{0,\tau_i} + \beta_{\text{att},\tau_i}\ell + \beta_{d,\tau_i}(\underline{s} - \bar{s}_{i\ell}) + e_i}}. \quad (6)$$

In Equation (6), $\bar{s}_{i\ell} \equiv \frac{1}{\ell} \sum_{k=1}^{\ell} s_{ik}$ is the raw average score up to attempt ℓ , which we use to proxy for posterior quality. Retaking probabilities depend on the gap between $\bar{s}_{i\ell}$ and the threshold and on the number of attempts, ℓ , reflecting the decline in gains from test-taking ability over attempts. They also depend on physicians' type, τ_i , and on unobserved systematic differences across individuals, $e_i \sim N(0, \sigma_{r,\tau_i}^2)$, whose variance is type-specific. As such, although reduced-form in nature, our retake model captures the main drivers of retaking decisions and allows us to endogenize the mass of test-takers in the counterfactuals.³⁷

Posterior Quality Given physician i 's score history $\vec{s}_i$, type τ_i , and entry cohort t_i^0 , we define her posterior quality as

$$\hat{\theta}_i \equiv \mathbb{E}[\theta_i \mid \vec{s}_i, \tau_i, t_i^0].$$

As shown in Appendix G, this posterior quality is the multi-attempt analog of the standard signal-to-noise-ratio, and is a weighted average of the prior means and the physician's attempt-adjusted average score. Within each mixture component, the weight on the average score rises with the number of attempts and falls with score noise.³⁸

who failed the exam after attempt ℓ , since they are more likely to have received a negative shock to their score. As a consequence, score gains between two consecutive attempts are higher than the causal effect of retaking.

³⁶We do not model the decision to take the exam for the first time because marginal changes in the threshold are unlikely to change an individual's career decisions or whether to migrate to Chile.

³⁷See (Krishna et al., 2018) for a microfounded dynamic model of retaking in a similar setting.

³⁸Table A.8 reports 2SLS estimates that replace the average initial score $\bar{s}_{jt}$ in Equation (5) with the hospital average of posterior quality $\hat{\theta}_i$, the model-consistent measure of latent quality among licensed physicians. A second specification lets prior means vary by region of training, the dimension our instruments exploit. Both yield output

Labor Market Matching Our goal is to understand how changes in the licensing threshold affect the allocation of physicians across hospitals. A central challenge in this task is that equilibrium matching probabilities may change under a counterfactual threshold. For example, raising the threshold reduces the mass of lower-quality physicians seeking jobs, potentially altering the degree of competition in the labor market.

We adopt a reduced-form modeling approach that flexibly captures the determinants of physician-hospital matches, and restrict our counterfactuals to *local* changes in order to reduce the sensitivity of the results to our statistical approximations (Chetty, 2009). Consider a physician i with posterior quality $\hat{\theta}_i$. We define the conditional matching probability (CMP) with location j in period t by:

$$CMP_{ijt} \equiv CMP\left([x_{i1t}, \dots, x_{i|\mathcal{J}|t}], \hat{\theta}_i, \tau_i, \mathbf{M}_{it}(\underline{s}), \kappa_t, j\right) = \frac{e^{v(x_{ijt}, \hat{\theta}_i) + g(\mathbf{M}_{it}(\underline{s}), \kappa_t | j)}}{1 + \sum_{j'} e^{v(x_{ij't}, \hat{\theta}_i) + g(\mathbf{M}_{it}(\underline{s}), \kappa_t | j')}}. \quad (8)$$

The function $v(\cdot)$ captures the determinants of matching probabilities based on physicians' and hospitals' preferences, which depend on the posterior quality and other match-specific factors such as distance between physicians and hospitals' locations, and amenities, included in $x_{i1t}, \dots, x_{i|\mathcal{J}|t}$. The function $g(\cdot)$ captures market-level forces—specifically, the extent of competition physician i faces when seeking a position at hospital j , which is determined by the mass of similarly or more qualified job seekers, $\mathbf{M}_{it}(\underline{s})$, and the vector of vacancies κ_t . Through this term, the model allows matching probabilities to respond endogenously to changes in the licensing threshold.

Although the matching probability is specified in reduced form, it is grounded in a microfounded model developed in Appendix F. In the model, hospitals have fixed capacities and vertical preferences over physicians' expected qualities, and the equilibrium assignment satisfies pairwise stability.³⁹ Under those assumptions, the labor market clears with a set of hospital-specific quality cutoffs that determine the minimum quality of physicians that can match with each hospital (Azevedo and Leshno, 2016).⁴⁰ A key implication is that the choice set of a physician with posterior quality $\hat{\theta}_i$ depends on the mass of physicians with similar or higher quality. That is, the equilibrium allocation is such that only competition from equally or better-ranked peers can affect a physician's match.

elasticities close to those in Table I.

³⁹Agarwal (2015) proposes and estimates an empirical model for the US centralized medical residency program under a similar set of assumptions.

⁴⁰This setup is motivated by two facts of our setting. First, wages are rigid: they are set by the public-sector wage schedule and have remained unchanged despite large supply shocks (see Appendix Figure A.2). Second, observed sorting patterns are consistent with vertical preferences: over 85% of the variance in physician quality is explained by differences between hospitals rather than within.

6.2 Estimation and Results

Latent Quality Table II, Panel A, presents maximum likelihood estimation results for the re-taking model specified in Equation (6). As expected, the probability of retaking decreases with the number of attempts and the gap between the average prior score and the cutoff. For example, foreign-trained physicians who fail the exam retake it with an 85% probability. The retake probability decreases by 0.57 percentage points for each point below the cutoff, while each additional attempt lowers the probability of retaking by 6.6 percentage points.

We estimate the model of scores via the simulated method of moments (SMM), using the estimated retaking probabilities and matching—separately for nationally- and foreign-trained physicians—the means and variances of scores over the first two attempts, the covariance between attempts for retakers, the mean gains over the first four attempts, and the cohort-specific first-attempt score distribution. Appendix Table A.9 and Appendix Figure A.3 show that the model reproduces the first-attempt distributions and targeted moments well. Panel B of Table II reports the implied cohort-specific quality moments and signal-to-noise ratios, while Table A.3 reports the remaining structural parameters. Foreign-trained physicians have lower and more dispersed latent quality than nationals and noisier scores.⁴¹ As a result, the exam’s signal-to-noise ratio is high and similar across the two groups on average, but evolves differently across cohorts: it stays roughly constant for nationals (0.71 to 0.73 between 2013 and 2018) while declining for foreigners (0.77 to 0.65) as their measurement error rises over the migration wave. The estimated first gain in test-taking ability is $\hat{\gamma} = 12.67$, below the raw average score increase between attempts (Appendix Table A.9), reflecting that the raw difference overstates the causal gain (Gilraine and Penney, 2025). Score gains decay at an exponential rate of $\hat{\rho} = 0.41$ —i.e., they decrease by roughly 34% between two consecutive attempts.

Three features of the data identify the model of scores. First, we observe the first-attempt score of every test-taker, so the cohort-specific first-attempt distributions identify the distribution of $\theta_i + \epsilon_{i1}$ by type and entry cohort. Because the noise is normal, the shape of these distributions determines the mixture weights and component means of latent quality. Second, repeated attempts allow us to separate the score variance into latent quality and noise. We assume that the individual effect e_i in Equation (6) is independent of θ_i and of the noise in future attempts, so that retaking depends on latent quality only through past scores. Under this assumption, the covariance between first and second scores among retakers measures how much of the first score persists, which identifies

⁴¹The higher variance of quality is expected, since foreign-trained physicians represent a more heterogeneous group in terms of experience, region of training, etc. The higher noise among foreign-trained physicians is likely explained by the fact that they have had fewer opportunities to practice for the test than nationals.

the signal-to-noise ratio, and the mean score gain of retakers, net of the implied regression to the mean, identifies γ . Third, the mean gains between the second and third attempts and between the third and fourth attempts identify ρ .

With the estimated parameters and given the history of score realizations for each physician, we use Equation (7) to estimate each individual's posterior quality.

Matching Probabilities We estimate the conditional matching probabilities by maximum likelihood following Equation (8). Locations denoted by j include all 29 Health Services and the outside option. We specify $v(x_{ijt})$ to include a rich set of alternative-specific time fixed effects, and alternative-specific shifters based on the physician's posterior quality, $\hat{\theta}_i$, an indicator for whether the physician is national, National_i , and an indicator for whether the physician is a specialist, Specialist_i . We also include the distance between the university of training and the centroid of location j for nationals, $\text{Distance}_{ij} \times \text{National}_i$. Finally, and consistent with our shift-share IV strategy, we include the percentage of physicians from i ' region of training $c(i)$ who worked at location j in $t - 1$; $\text{Share Physicians}_{c(i)jt-1}$. Thus, v_{ijt} is defined as follows:

$$v_{ijt} = \alpha^d \text{Distance}_{ij} \times \text{National}_i + \alpha^h \text{Share Physicians}_{c(i)jt-1} + \alpha_{jt} + \alpha_j^n \text{National}_i + \alpha_j^q \hat{\theta}_i + \alpha_j^{nq} \hat{\theta}_i \times \text{National}_i + \alpha_j^s \text{Specialist}_i. \quad (9)$$

The coefficients α_{jt} represent mean alternative-specific year fixed effects, and α_j^n , α_j^q , α_j^{nq} , and α_j^s allow those mean effects to vary with physicians' characteristics. We normalize $v_{ijt} = 0$ for the outside option.

As specified in Equation (8), the matching probabilities depend on the mass of physicians who pass the licensing exam and the vacancies in each location. For our empirical application, we capture those general equilibrium effects by discretizing quality into quality ranges for the computation of masses, and by specifying a simple linear function of the ratio between job seekers and vacancies (i.e., the inverse of "labor market tightness"):

$$g(M_{it}(\underline{s}), \kappa_t | j) = \beta_j' \frac{\mathbf{M}_{it}(\underline{s})}{\kappa_{jt}}, \quad (10)$$

where β_j is a 2×1 vector of location-specific coefficients and $\mathbf{M}_{it}(\underline{s}) = [M_{it}^0(\underline{s}), M_{it}^+(\underline{s})]'$ is a 2×1 vector composed of the mass of physicians in i 's quality range, $M_{it}^0(\underline{s})$, and the mass of physicians above i 's quality range, M_{it}^+ .⁴² We use four (posterior) quality quantiles to construct $\mathbf{M}_{it}$. We

⁴²Let m_t be the number of test-takers in period t and $h_t(s, X)$ the joint distribution of scores and characteristics

proxy the number of vacancies κ_{jt} with the ratio of beds and the stock of physicians in the previous period; that is, $\tilde{\kappa}_{jt} = \frac{\text{Beds}_{jt}}{\text{Stock of Physicians}_{j,t-1}}$.

The coefficients β_j determine the extent to which the matching probabilities change with counterfactual thresholds. As Equation (9) includes alternative-specific year fixed effects and alternative-specific coefficients on posterior quality, β_j is identified from changes over time in the mass of licensed physicians in each quality range, interacted with location-specific vacancies. These changes come from the entry of foreign-trained physicians and the growth of domestic medical schools (Section 3). The identifying assumption is that they are unrelated to unobserved location-specific shocks to the matching of physicians in a given quality range.

Results are presented in Table III. Column (1) shows the estimates from a specification in which $g(\cdot)$ depends only on other physicians within the same quality range (M_{it}^0). Column (2) allows $g(\cdot)$ to also depend on the mass of physicians with a higher quality range (M_{it}^+). Column (3) allows the effects of these masses to vary across three different hospital-quality tiers, which we construct based on the average quality of their physicians. We do not find significant heterogeneity along that dimension. As a placebo check, in Column (4) we show that the mass of physicians in a lower quality range (which we denote M_{it}^-) does not affect the matching probabilities. Across specifications, the matching probabilities depend negatively on distance. Also, and in line with our shift-share IV strategy, the matching probabilities depend positively on $\text{Share Physicians}_{c(i)jt-1}$. As predicted by the theory, we find that matching probabilities depend negatively on the (inverse of) labor market tightness. The effects are mostly concentrated within the same quality range (M_{it}^0), with a much smaller, positive effect from quality ranges above (M_{it}^+).

Input Elasticities The matching probabilities allow us to compute the elasticity of the quantity and quality of physicians in each hospital with respect to the licensing threshold. We compute these elasticities numerically, by simulating the hospital-physician matches under the baseline licensing threshold $\underline{s}$ and under a counterfactual threshold $\underline{s}' = \underline{s} + \Delta\underline{s}$, and compute the resulting qualities and quantities and their elasticities. We use $\Delta\underline{s} = 5$. In Appendix I, we show that the resulting simulated changes in quantity and quality lie mostly within the convex hull of the variation

in period t . With a slight abuse of notation, we define the mass of physicians who pass the exam in year t and have expected quality within range r :

$$M_{r,t}(\underline{s}) \equiv m_t \int_X \int_{s > \underline{s}: \hat{\theta}(s) \in r} h_t(s, X) ds dX. \quad (11)$$

The mass of physicians in i 's quality range is $M_{it}^0(\underline{s}) = M_{r(\hat{\theta}_i),t}(\underline{s})$ and the mass of physicians above i 's quality range, $M_{it}^+ = \sum_{r' > r(\hat{\theta}_i)} M_{r',t}$. Notice that, for physicians in the lowest quality range, only M_{it}^0 depends on the licensing score, as the licensing score only affects the mass at the bottom of the distribution.

identifying the production function.⁴³

These matching probabilities imply that the average quantity elasticity, pooling across years and hospitals, is -0.095, and the average quality semi-elasticity is 0.353 standard deviations. The ratio of these equals -3.7, which implies that, on average, a 10% decrease in labor due to a higher threshold would increase quality by 0.37 standard deviations.

The flexible specification for the matching probabilities allows for rich heterogeneity in the elasticities across hospitals. Appendix Table A.10 presents the results from estimating a series of OLS regressions in which we project the labor elasticity (Columns 1-4) and the quality semi-elasticity (Columns 5-8) on hospital observables. Consistent with our assumptions on vertical preferences, our estimates predict that increasing the licensing threshold has a larger effect on hospitals with lower-quality doctors (below-median average scores). We also find that increasing the licensing threshold would have higher impacts on hospitals with more scarcity (below-median ratio of physicians per patient) and in the north (above-median latitude).

7 The Impact of Increasing the Licensing Threshold

Taken together, the input and output elasticities estimated in the previous sections enable us to measure the elasticities of hospital-specific health outcomes to changes in the licensing threshold.

We estimate hospital-specific input elasticities that we then aggregate. We can restate Equation (1) and express the elasticity of the aggregate health outcome across hospitals with respect to the licensing threshold as

$$\eta_{\underline{s}}^{Y,T} = \underbrace{\alpha_L \cdot \bar{\eta}_{\underline{s}}^{L,T}}_{\text{Licensing Quantity Effect}} + \underbrace{\alpha_{\theta} \cdot \bar{\bar{\eta}}_{\underline{s}}^{\bar{\theta},T}}_{\text{Licensing Quality Effect}},$$

where $\bar{\eta}_{\underline{s}}^{L,T} = \sum_{j \in \mathcal{J}} p_j \cdot \eta_{\underline{s}}^{L_j,T}$ is the average elasticity of labor in period T with respect to $\underline{s}$, $\bar{\bar{\eta}}_{\underline{s}}^{\bar{\theta},T} = \sum_{j \in \mathcal{J}} p_j \cdot \bar{\eta}_{\underline{s}}^{\bar{\theta}_j,T}$ is the average semi-elasticity of quality in period T with respect to $\underline{s}$, and p_j are hospital-specific Pareto weights.⁴⁴ The index T denotes the time horizon for evaluating the

⁴³Due to sparsity, we estimate these matching probabilities at the Health-Service level. We transform the matching probabilities into hospital-specific elasticities by assigning physicians to hospitals in proportion to their relative size within the health service. We provide more details on how we perform this calculation in Appendix H.

⁴⁴These average elasticities are consistent with a planner considering aggregate health outcomes of the form:

$$Y_T \equiv \prod_{j \in \mathcal{J}} y_{jT}^{p_j}.$$

The main appeal of this aggregation is that, when $p_j > 0 \forall j$, it penalizes solutions where the outcome is very low for

policy. The option to retake the exam creates a distinction between the immediate and future impacts of increasing the licensing threshold, due to fact that test-takers in any period t' include those retaking the exam after failing in an earlier period $t < t'$. As such, increasing the threshold in a period t_0 not only impacts who passes in period t_0 but also the set of test-takers in any period $t' > t_0$.

In the next subsections, we evaluate the effects of increasing the licensing threshold over two time horizons. The “short-run” effects ($T = 0$) occur in the period of the change, while the “medium-run” effects ($T > 0$) occur in later periods when the increase is permanent.

7.1 Short-run Impacts of Increasing the Licensing Threshold

Figure VI plots, for each hospital, the short-run elasticity of quantity and semi-elasticity of quality with respect to the licensing threshold in 2018 ($T = 0$). The size of the marker is proportional to the number of patients in each hospital, which we use as Pareto weights to compute the average elasticities shown in the solid diamond. The average labor elasticity is $\bar{\eta}_s^{L,0} = -0.082$ and the average quality semi-elasticity is $\bar{\eta}_s^{\bar{\theta},0} = 0.35$, with quality measured in standard deviations. The resulting aggregate marginal rate of transformation is $\bar{R}(\underline{s}) \equiv -\bar{\eta}_s^{\bar{\theta},0} / \bar{\eta}_s^{L,0} = 4.3$.

The dashed black line in Figure VI corresponds to the “iso-mortality curve,” which depicts the quantity elasticity needed to keep mortality constant, given the quality semi-elasticity. The inverse of the slope of this curve is $-\hat{\alpha}_L^{\text{mortality}} / \hat{\alpha}_{\bar{\theta}}^{\text{mortality}} = -3.85$, as estimated in Section 5.3.

We find that the marginal rate of transformation is close to the marginal rate of technical substitution for the case of mortality, i.e., $\bar{R}(\underline{s}) \approx \hat{\alpha}_L^{\text{mortality}} / \hat{\alpha}_{\bar{\theta}}^{\text{mortality}}$. In other words, the increase in quality offsets the decrease in quantity so that, on net, the elasticity of mortality with respect to the licensing threshold is small and statistically indistinguishable from zero.

Time-series Evolution of Short-run Impacts We analyze how the effect of the licensing policy depends on the market-level fundamentals affected by the migration wave: the mass and quality distribution of physicians—both among test-takers and among the stock of physicians employed in hospitals—and the degree of labor-market tightness. Specifically, we compare the policy effects found for 2018—5 years after the onset of the migration wave—with those in earlier years.

Panel A in Figure VII shows the evolution of short-run input elasticities over time. The negative quantity elasticity (in red) follows a U-shaped pattern. It is closest to zero in 2013, when there

some hospitals.

are few physicians at the margin of passing (see Table A.1). In the following years, the elasticity becomes more negative through 2017 as migration increases the number of physicians close to the threshold, then moves back toward zero in 2018. Even if the number of marginal physicians increases throughout the sample period, more physicians had already been hired when later cohorts arrived. This attenuates the elasticity mechanically since the baseline is higher, but also through the lower remaining capacity in the public sector which increases the share of physicians who match with the outside option as captured by the labor matching function. The positive quality semi-elasticity follows an inverted-U pattern, rising through 2017 and then falling in 2018.

Panel B in Figure VII shows the short-run effects on mortality and access to care for all years in our sample. The mortality effects are small and statistically indistinguishable from zero in every year, because the quality gains of increasing the threshold roughly offset the quantity reductions. Access to care would decrease regardless of the year in which the policy is enacted. For 2018, a 10% increase in the licensing threshold would change mortality by -0.07% and would decrease access by 0.9%. The evolution of the magnitudes of these effects mirrors the elasticities of inputs described in Panel A. The service-rate effect is largest in 2017, when a 10% increase in the licensing threshold would change mortality by -0.09% and would decrease the access by 1.9%.⁴⁵

Evaluating the policy across different years provides a form of external validity (List, 2020). The magnitudes of the effects align with our theoretical framework: in years with looser labor markets and lower scarcity (post-migration), the impacts of changing the threshold are more muted, while in years with greater scarcity (mid-migration wave, 2017), the same policy change has larger effects on inputs and access to care. This pattern—where the optimal stringency depends on market-level fundamentals that characterize the quantity-quality trade-off—strengthens confidence that our framework can inform licensing policy in other settings with different labor market conditions.

The Role of Sorting and the General-Equilibrium Effects Absent any sorting, the marginal rate of transformation effect of quality depends on the difference between the average quality of the supra-marginal individuals and the quality of marginal individuals.

To quantify the impact of the general equilibrium effects in our counterfactuals, we reproduce the main results under the assumption that the matching probabilities do not change with changes in the counterfactual threshold. Table A.11 shows the results. Generally, omitting GE effects would result in quantity and quality elasticities that are similar but slightly lower in magnitude

⁴⁵As noted above, the effects of the policy are mediated by the fraction of physicians that match with the inside option. Therefore, to broaden the external validity of our results, we consider a counterfactual scenario where all physicians match with the inside option. In that case, the elasticities would be substantially higher: for 2018, a 10% increase in the threshold would change mortality by -0.15% and lower access by 4.2%.

than our main estimates (comparing columns (1) and (2), and columns (3) and (4)). However, the magnitude of $\bar{R}(\underline{s})$ —which determines the sign of the effects on mortality—remains virtually unchanged. Overall, the lower predicted effects on quantity are offset by lower predicted effects on quality. The magnitudes of the estimated net impact of the reform on mortality and access are similar as well.

7.2 Medium-run Impacts of Increasing the Licensing Threshold

We now turn to analyzing the dynamic effects of permanently changing the licensing threshold. The critical mechanism we stress in this subsection is that retaking might mitigate the relevance of the licensing threshold as test-takers improve their scores over attempts. For instance, 82% of the test-takers who failed the licensing exam on their first attempt in 2013 had passed it by 2018.

To quantify the differences between the short-run and medium-run effects of changing the threshold, we leverage the estimated model of test scores and retaking behavior described in Section 6. We compute the medium-run impacts by forward-simulating the model under the status quo ($\underline{s} = 51$), and compare it to a counterfactual scenario in which we set the threshold *permanently* higher, starting in 2013.⁴⁶ Comparing outcomes across these scenarios for any year $t \geq 2013$ within our sample provides the “medium-run” elasticities.⁴⁷

Table IV shows the simulated passing rates for the 2013 cohort, under the current threshold ($\underline{s} = 51$) and under a counterfactual higher threshold ($\underline{s}' = 56$). In the status quo, 76.3% of test-takers pass on their first attempt (in 2013), compared with 68.0% under the counterfactual policy. The 8.3 percentage points gap in passing rates determines the short-run elasticities. However, this gap narrows over time as individuals retake the exam. By 2018, 93.0% of physicians from the 2013 cohort pass under the status quo, compared with 89.6% under the counterfactual, reducing the gap to 3.4 percentage points. Panel A of Appendix Figure A.4 illustrates the divergence between the short-run and the medium-run elasticity over time: while (by construction) short- and medium-run elasticities are identical at the time of implementation (2013), medium-run elasticities fall to roughly 46% (quantity) and 54% (quality) of their short-run values by 2018, as some marginal test-takers who initially fail eventually pass in subsequent years. Despite this attenuation, increasing the threshold in 2013 still generates meaningful changes in quantity and average physician quality in the following years. Still, on net, the effect on mortality remains small and statistically insignificant

⁴⁶As in the previous section, we estimate the numerical elasticities with $\bar{s} = 56$, and divide the results by 5.

⁴⁷In all these simulations we account for the fact that matching probabilities change endogenously with the threshold. Matching probabilities in any year t depend on the mass of test-takers who pass the exam through the term $\mathbf{M}_{it}(\underline{s})$ as well as the shares of physicians from each country through the term $\text{Share Physicians}_{c(i)j,t-1}$. Our simulations endogenize these variables.

in the medium run (Appendix Figure A.4, Panel B).

8 Conclusion

Occupational licensing is a widely used regulation in labor markets to ensure a minimum quality standard. However, it comes at the cost of reducing the labor supply. This quantity-quality trade-off is particularly relevant in healthcare, where licensing is ubiquitous and its design may have first-order implications. In this paper, we first characterize this trade-off with a simple theoretical framework. Guided by the theory, we develop and estimate an empirical model to quantify the quantity-quality trade-off in physician licensing in Chile, where licensing is decided based on a minimum passing score on an exam.

Our empirical strategy is built around estimating two sets of elasticities needed to evaluate the quantity-quality trade-off: the elasticities of inputs—the quantity and quality of labor—with respect to the licensing threshold, and the elasticities of hospital outputs with respect to those inputs. We begin by estimating production functions to provide novel results for the elasticity of hospital outcomes with respect to the quantity and quality of physicians. Key to our analysis, we leverage quasi-experimental variation in these inputs arising primarily from an unprecedentedly large immigration of physicians to Chile. We find that both the quantity and the quality of physicians affect healthcare outcomes. As a consequence, changing the licensing threshold generally involves a quantity-quality trade-off.

To estimate the elasticities of inputs, we infer the latent quality of physicians from their individual exam score histories and introduce a microfounded labor matching function to model the sorting of physicians with different quality levels across hospitals under alternative licensing scenarios. Through a counterfactual analysis that raises the licensing threshold, we find that the reduction in mortality from improving physician quality offsets the increase in mortality due to the reduced physician quantity. On net, the effect on mortality is statistically indistinguishable from zero, while access falls.

These findings suggest that physician licensing would reduce the number of available physicians and raise their average quality without improving the quality of care. Comparing adjustments to licensing thresholds with alternative strategies—such as expanding access to high-quality medical education, which may increase the physician supply without compromising quality—offers a promising avenue for future research.

Although our quantitative results are specific to our context, our model and empirical strategy

offer novel estimates for the role of quantity and quality on healthcare outcomes, and provide a rich framework for evaluating the stringency of licensing policies in other settings.

Our focus on local counterfactual analysis aligns closely with how policy changes are actually implemented. Chilean policymakers are considering adjustments near the current threshold to preserve system stability and minimize disruption. Importantly, our framework is well-suited for guiding progressive policy reforms. Rather than requiring large one-time changes, policymakers can implement modest threshold adjustments (e.g., 1-2 points), collect new data, and use our framework to reassess optimality before deciding on subsequent adjustments. This iterative approach allows for course correction and enables our methodology to inform sequential policy optimization over time, making it particularly valuable for practical policy design.

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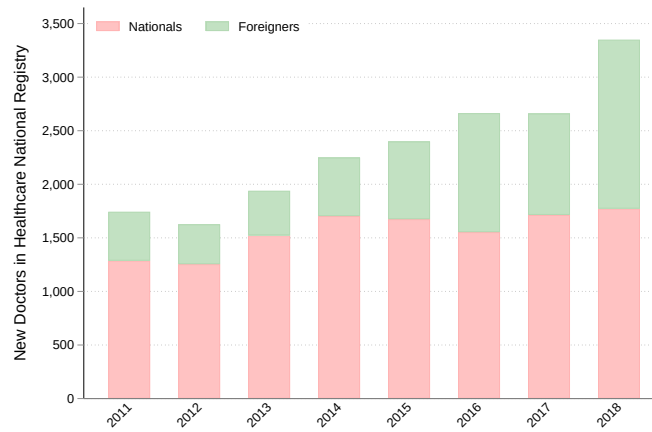
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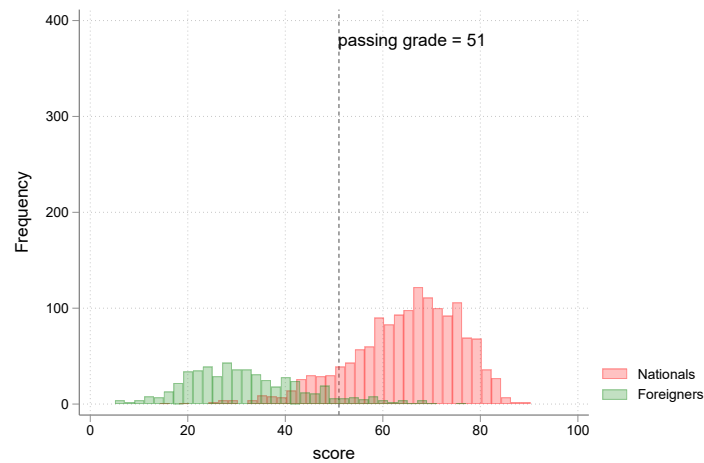
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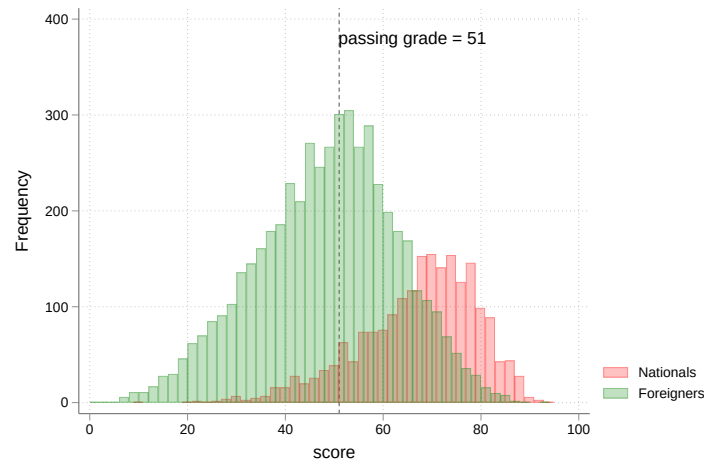
Figure I: Physicians' Labor Market



A. Registered Physicians



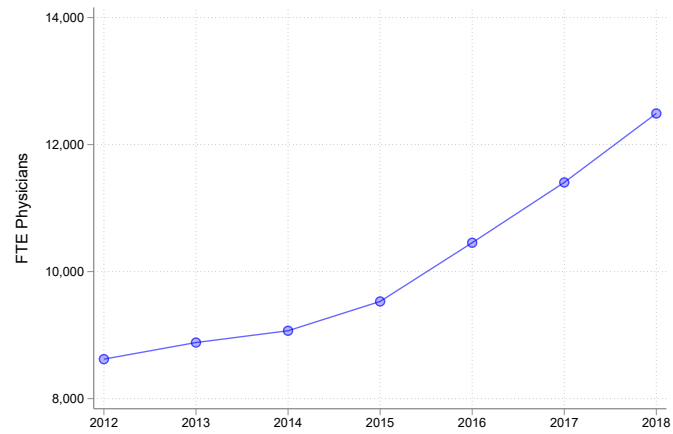
B. Scores in 2013



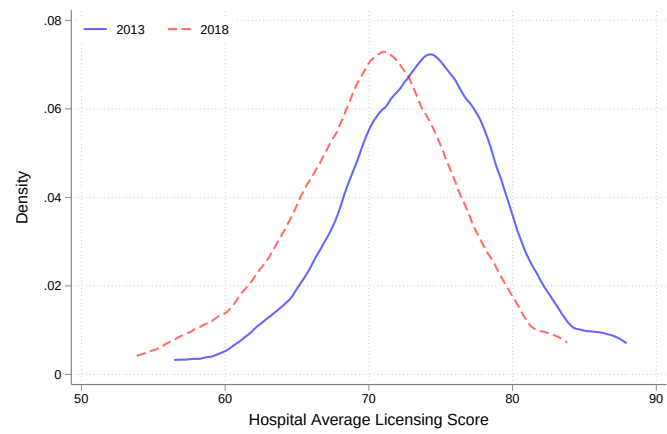
C. Scores in 2018

Notes: Panel A shows the number of newly registered physicians in the National Registry of Healthcare Providers between 2011 and 2018 ([RNPI, 2024](#)). The bars are divided into locally trained physicians (red) and foreign-trained physicians (green). Panels B and C display the distribution of test scores from [ASOFAMECH \(2019\)](#) for locally trained (red) and foreign-trained (green) physicians in 2013 and 2018, respectively. The dashed line in both graphs represents the minimum passing score.

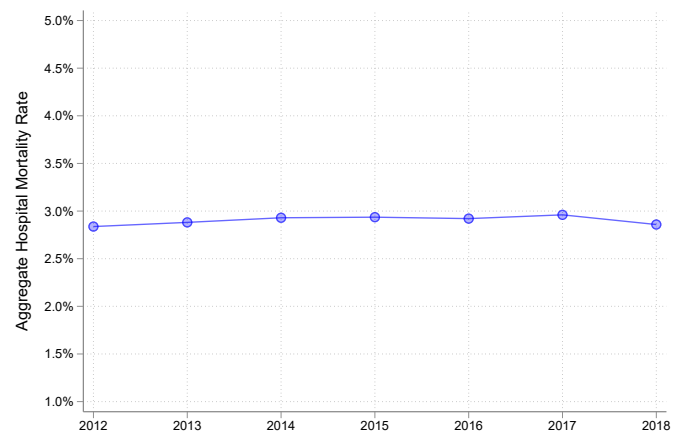
Figure II: Physicians in Public Hospitals



A. Quantity



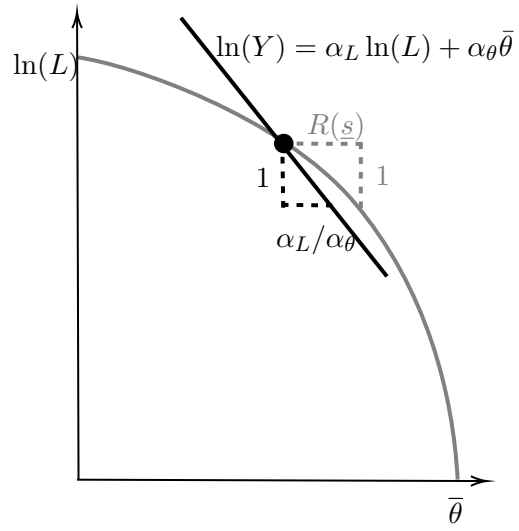
B. Scores



C. Hospital Mortality

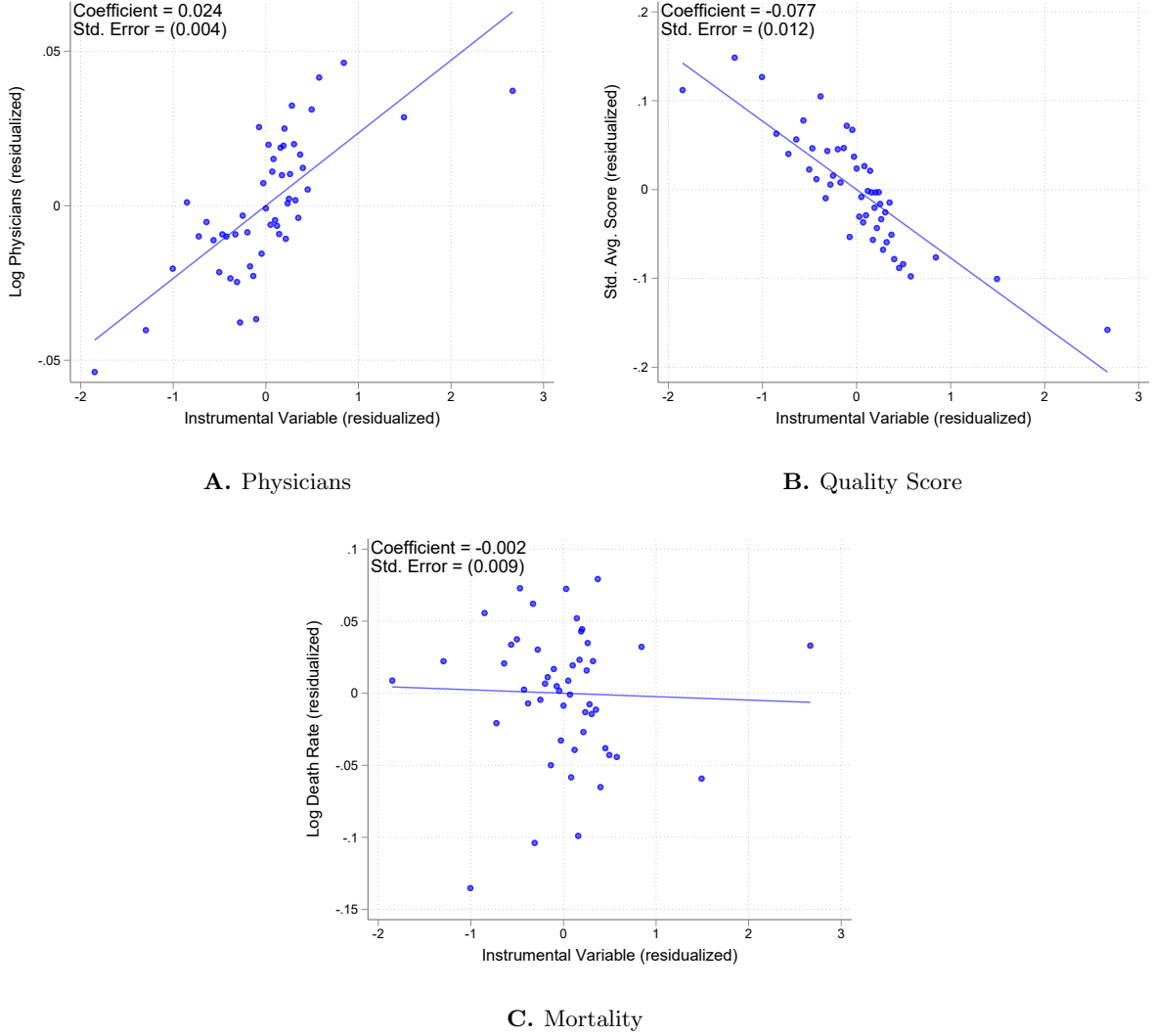
Notes: Panel A plots the number of full-time equivalent physicians working in public hospitals using data from [SIRH \(2019\)](#). Panel B presents the distribution of public hospitals' average licensing scores of physicians in 2013 and 2018 using data from [ASOFAMECH \(2019\)](#). Panel C depicts the average aggregate mortality rate in public hospitals using data from [DEIS \(2019\)](#).

Figure III: Simple Licensing Problem



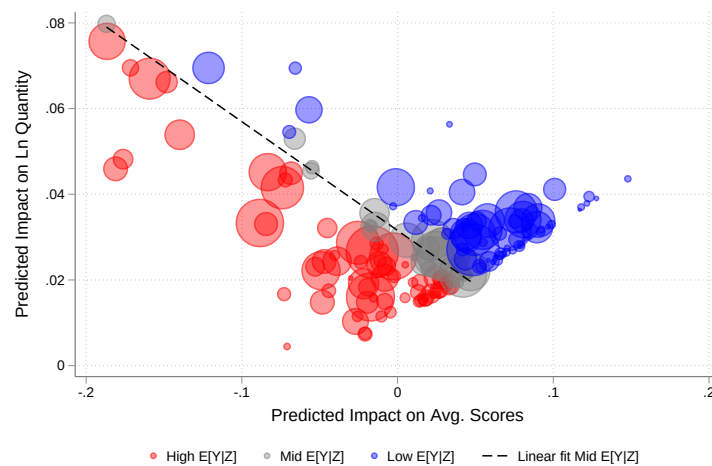
Notes: The figure depicts the licensing problem and the quantity-quality trade-off. The gray curve represents the set of possible combinations between labor and quality that are feasible. When the threshold is increased from the point depicted by the dot, decreasing quantity by one log point implies an increase in quality by $R(\underline{s})$. The solid black line represents the iso-outcome curve. The slope depends on the marginal product of the inputs. Increasing the licensing threshold would improve outcomes if $R(\underline{s}) > \alpha_L/\alpha_\theta$.

Figure IV: Effects of the Labor Supply Shock on Inputs and Mortality



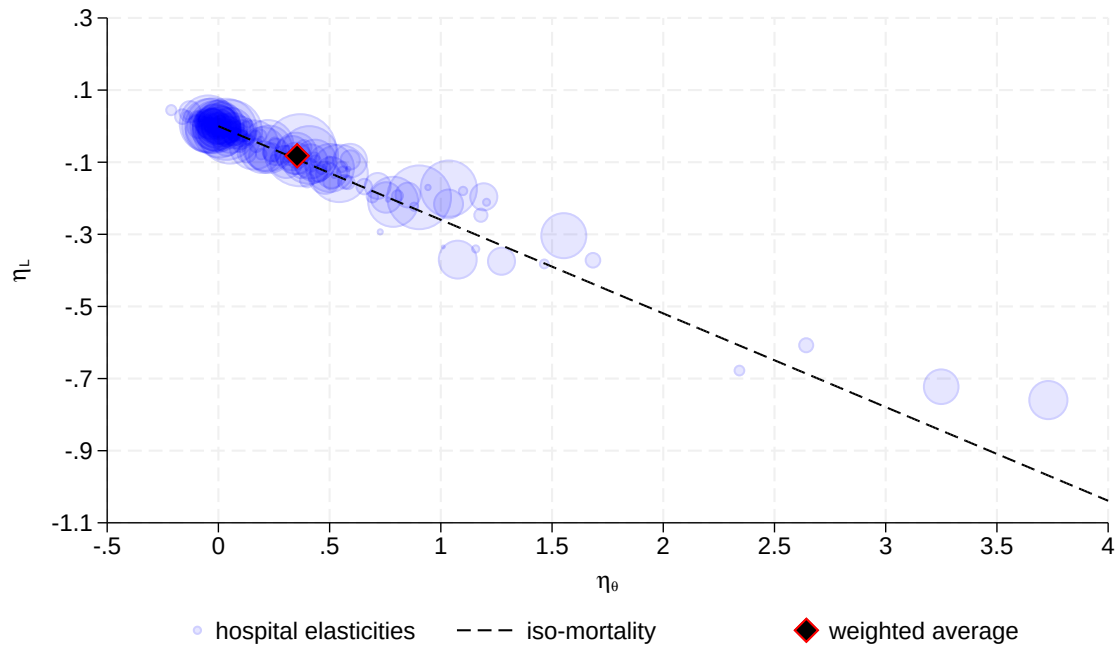
Notes: This figure presents binned scatterplots of the log number of physicians at each hospital (Panel A), their average test scores (Panel B), and the log hospital death rate (Panel C) against the standardized instrumental variable. We residualize each outcome and the instrument with respect to the controls and fixed effects included in Equation (5). Each point plots the mean residualized outcome against the mean residualized instrument within one of fifty bins. Solid lines show linear fits estimated using the underlying observations. Each panel reports the coefficient on the standardized instrument and its standard error from the corresponding full regression. We standardize the instrument before residualization, so coefficients measure changes in outcomes associated with a one-standard-deviation increase in the instrument.

Figure V: Instrument-predicted Physician Quantity and Quality by Hospital



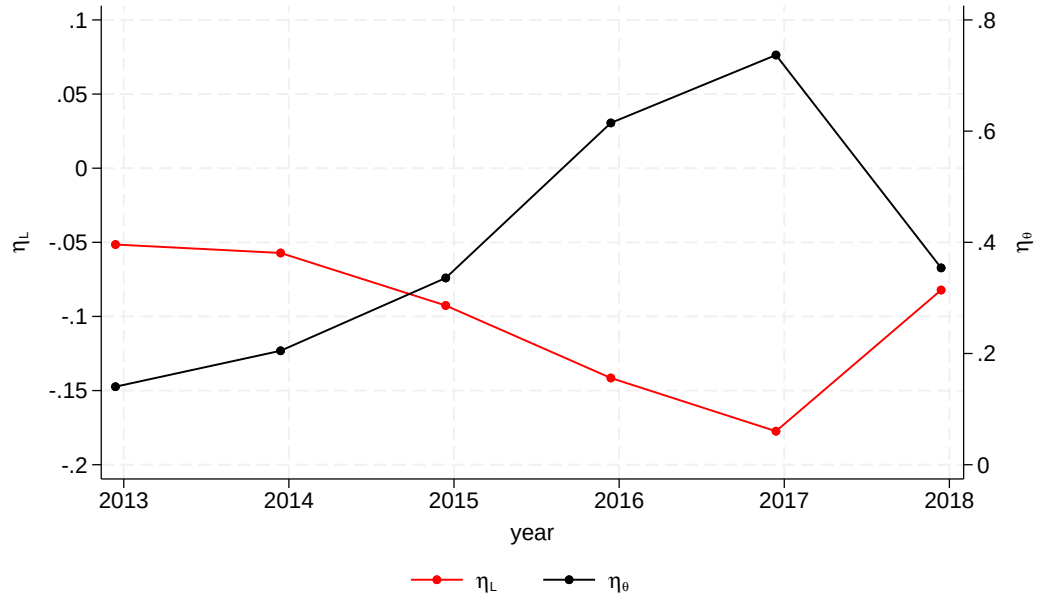
Notes: These figures plot the instrument-predicted quantity and quality of physicians for each hospital. Predictions are obtained by combining the first-stage coefficients with average instrument values at the hospital level, isolating only the variation in inputs identified by the instruments. Marker sizes are proportional to baseline hospital size. Hospitals are color-coded into terciles based on their predicted inpatient mortality. Predicted inpatient mortality is obtained by combining the reduced-form coefficients with average instrument values at the hospital level, isolating only the variation in outputs identified by the instruments. The figure also displays the linear relationship between predicted physician quantity and quality for hospitals whose predicted outcome is close to the sample mean.

Figure VI: Short-run Elasticity of Quantity and Semi-elasticity of Quality by Hospital in 2018

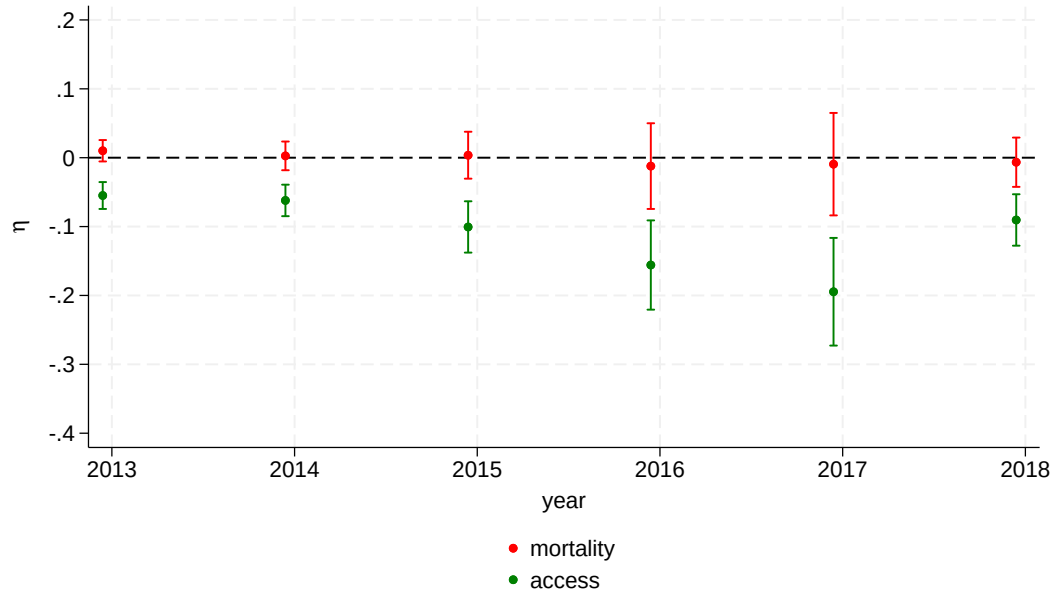


Notes: This figure shows a scatter plot with estimates for the elasticity of quantity and the semi-elasticity of quality with respect to the licensing threshold. Each dot represents a hospital. The size of the marker is proportional to the number of patients. Combinations of elasticities above the dashed line are such that mortality decreases when the threshold is increased.

Figure VII: Time Series of Short-run Elasticities



A. Labor Quantity and Quality



B. Hospital Outcomes

Notes: Panel A shows the evolution of the average short-run elasticities of the quantity and quality of physicians with respect to the licensing threshold. Panel B shows the resulting average effects on patient mortality and access; the bars are 95% confidence intervals obtained by the delta method from the exposure-robust standard errors of the production-function estimates in Table I.

Table I: Impact of Physicians' Quantity and Quality on Patient Outcomes

	Discharges	Mortality					Complications
	Ln Inpatient Admissions	Ln death rate In-Hospital	Ln death rate 28-Days	Ln Actual over Pred. death rate	Ln death rate Amenable	Ln death rate Non-amenable	Ln readmission rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ln Physicians ($\hat{\alpha}_L$)	1.01 (0.31)	-0.67 (0.20)	-0.70 (0.23)	-0.65 (0.16)	-0.93 (0.30)	-0.35 (0.26)	-0.55 (0.16)
Std. Physicians' Score ($\hat{\alpha}_\theta$)	-0.02 (0.09)	-0.17 (0.07)	-0.20 (0.08)	-0.21 (0.06)	-0.25 (0.09)	-0.09 (0.08)	-0.20 (0.06)
Observations	1,404	1,404	1,404	1,404	1,379	1,396	1,404
Mean Dep. Var. (levels)	5,664	3.32	5.22	0.75	10.47	2.11	3.47
F-stat (First-stage)	21.80	21.80	21.80	21.80	26.16	20.35	21.80
Anderson–Rubin (χ^2) p-value	0.00	0.01	0.00	0.00	0.00	0.47	0.01

Notes: This table presents the impact of physician quantity and quality on public hospital performance. Estimates come from two-stage least-squares estimation of Equation (5). The set of conditions amenable to physician treatment is discussed in the main text. The mean dependent variable is presented in levels instead of logs. We present exposure-robust standard errors clustered at the region-of-origin, i.e., the shock level, in parentheses (Adão et al., 2019; Borusyak et al., 2022).

Table II: Model Estimates

Panel A: Model of Retaking		
	Foreign-trained	Nationally trained
β_{att}	-0.695 (0.052)	-0.433 (0.097)
β_d	-0.060 (0.005)	-0.093 (0.015)
β_0	4.597 (0.219)	2.745 (0.287)
$\ln(\sigma_r^2)$	0.637 (0.171)	-0.073 (0.453)
$\frac{\sigma_r^2}{\sigma_r^2 + \pi^2/3}$	0.365 (0.040)	0.220 (0.078)
Observations	6,731	1,114

Panel B: Model of Scores						
Implied quality moments and SNR, by cohort						
	2013	2014	2015	2016	2017	2018
<i>Foreigners</i>						
$\hat{\mu}_\theta$	35.7 (0.4)	33.7 (0.4)	42.1 (0.6)	43.1 (0.7)	44.5 (0.6)	49.1 (0.7)
$\hat{\sigma}_\theta$	12.1 (0.6)	13.7 (0.5)	13.2 (0.4)	14.5 (0.2)	14.1 (0.2)	13.1 (0.4)
$\hat{\sigma}_\epsilon$	6.6 (1.1)	7.1 (0.8)	7.6 (0.6)	8.2 (0.3)	8.9 (0.2)	9.6 (0.5)
SNR	0.77 (0.07)	0.79 (0.05)	0.75 (0.04)	0.76 (0.02)	0.72 (0.01)	0.65 (0.04)
Observations	345	434	711	1,658	1,937	2,188
<i>Natives</i>						
$\hat{\mu}_\theta$	65.1 (0.6)	66.2 (0.7)	67.2 (0.7)	69.4 (0.6)	67.6 (0.7)	68.5 (0.8)
$\hat{\sigma}_\theta$	9.4 (0.7)	9.7 (0.3)	9.2 (0.2)	9.5 (0.3)	8.9 (0.5)	8.9 (0.8)
$\hat{\sigma}_\epsilon$	5.9 (1.1)	5.6 (0.6)	5.5 (0.3)	5.4 (0.5)	5.3 (0.9)	5.3 (1.3)
SNR	0.71 (0.10)	0.74 (0.05)	0.74 (0.03)	0.75 (0.04)	0.73 (0.09)	0.73 (0.13)
Observations	1,438	1,578	1,634	1,611	1,658	1,806

Notes: This table reports estimates from the dynamic model of scores and retaking. Panel A presents maximum likelihood estimates of Equation (6) by physicians' region of training, using failed physician-attempts from the 2013–2018 entry cohorts. The coefficients β_{att} , β_d , and β_0 correspond to the number of attempts, the gap $\bar{s} - \bar{s}_{i\ell}$, and the intercept. The parameter σ_r^2 is the variance of the individual effect e_i , and $\sigma_r^2/(\sigma_r^2 + \pi^2/3)$ is the share of total latent variance due to between-physician heterogeneity. Panel B reports, by type and entry cohort, the implied mean and standard deviation of latent quality, $\hat{\mu}_\theta$ and $\hat{\sigma}_\theta$, the standard deviation of score noise, $\hat{\sigma}_\epsilon$, and $\text{SNR} \equiv \hat{\sigma}_\theta^2/(\hat{\sigma}_\theta^2 + \hat{\sigma}_\epsilon^2)$. Panel B observations are first-time test-takers. Its point estimates are bootstrap means, and its standard errors are standard deviations across 79 bootstrap iterations. Standard errors are reported in parentheses.

Table III: Conditional Matching Probabilities Estimates

	Alternative Models			Placebo
	(1)	(2)	(3)	(4)
Distance _{ij}	-0.233 (0.014)	-0.233 (0.014)	-0.233 (0.014)	-0.233 (0.014)
Share _{ijt-1}	0.595 (0.148)	0.593 (0.148)	0.595 (0.148)	0.598 (0.148)
(M _{it} ⁰)/κ _{jt}	-0.720 (0.151)	-0.850 (0.158)	-0.970 (0.189)	
(M _{it} ⁺)/κ _{jt}		0.050 (0.019)	0.065 (0.033)	
(M _{it} ⁰)/κ _{jt} × 1[r _j = 2]			0.022 (0.097)	
(M _{it} ⁰)/κ _{jt} × 1[r _j = 3]			0.152 (0.122)	
(M _{it} ⁺)/κ _{jt} × 1[r _j = 2]			-0.021 (0.038)	
(M _{it} ⁺)/κ _{jt} × 1[r _j = 3]			-0.007 (0.047)	
(M _{it} ⁻)/κ _{jt}				-0.007 (0.017)
Log likelihood	-15245.51	-15241.85	-15239.88	-15257.11

Notes: This table reports estimates from the multinomial logit model of physician matching across 29 Health Services and the outside option. The estimating sample has 428,790 physician-location observations (30 options for each of 14,293 physicians). Distance_{ij} is the distance between a national physician's university and location j , and Share_{ijt-1} is the share of physicians from i 's region of training who worked at location j in the previous year. M_{it}^0 , M_{it}^+ , and M_{it}^- denote the masses of eligible physicians in the same, higher, and lower quality ranges, respectively, and κ_{jt} denotes vacancies. Column (1) includes M_{it}^0/κ_{jt} ; Column (2) adds M_{it}^+/κ_{jt} ; Column (3) allows both coefficients to vary across hospital-quality tiers, with $1[r_j = k]$ indicating tier k ; and Column (4) replaces these terms with the placebo M_{it}^-/κ_{jt} . As shown in Equation (9), all specifications include alternative-specific year fixed effects and alternative-specific coefficients for nationality, posterior quality, their interaction, and specialty. Standard errors are reported in parentheses.

Table IV: Simulated Passing Year for 2013 Cohort

Year	$\underline{s} = 51$		$\underline{s} = 56$	
	Passing rate	Cumulative passing rate	Passing rate	Cumulative passing rate
2013	76.3	76.3	68.0	68.0
2014	8.4	84.7	12.0	80.0
2015	5.4	90.1	5.3	85.3
2016	1.9	92.0	2.5	87.8
2017	0.7	92.8	1.4	89.2
2018	0.3	93.0	0.4	89.6

Notes: The table shows the simulated passing rates over time for the cohort of physicians who take the exam for the first time in 2013, under a (baseline) scenario in which the threshold is kept at 51 points and under a counterfactual scenario in which the threshold is increased to 56 points.

ONLINE APPENDIX

Physicians' Occupational Licensing and the Quantity-Quality Trade-Off

Juan Pablo Atal, Tomás Larroucau, Pablo Muñoz, and Cristóbal Otero

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A Score Imputation

The EUNACOM exam was introduced in 2009. To account for the quality of physicians licensed before EUNACOM was introduced, we impute their hypothetical scores. This appendix outlines the imputation procedure.

Formally, let y_{ih} denote the licensing score of physician i in hospital h . Our baseline model for scores is

$$y_{ih} = \mu_h + \tilde{\alpha}_{r(i)}^h + \varepsilon_{ih},$$

where μ_h represents the hospital-level mean, $\alpha_{r(i)}^h$ captures the region-of-origin effect (or “differential score”) for physicians from region r working in hospital h , and ε_{ih} is an idiosyncratic error term with a mean of zero.

To improve out-of-sample predictions, which is particularly important when hospitals have very few physicians from a particular region, we apply an empirical Bayes shrinkage procedure (Efron and Morris, 1973; Walters, 2024). Specifically, we regress scores on region-of-origin fixed effects for each hospital h using OLS with sum-to-zero constraints (i.e., $\sum_r \alpha_r^h = 0; \forall h$). We treat each α_r^h as coming from a prior distribution centered at 0—which reflects the sum-to-zero constraint—with limiting variance τ_h^2 . Given the OLS estimate $\hat{\alpha}_r^h$ and its sampling variance $\widehat{\text{Var}}(\hat{\alpha}_r^h)$, the empirical Bayes estimate (posterior mean) is a weighted average of the OLS estimate and the prior mean (zero). Specifically:

$$\tilde{\alpha}_r^h = \underbrace{\frac{\tau_h^2}{\tau_h^2 + \widehat{\text{Var}}(\hat{\alpha}_r^h)}}_{\text{shrinkage factor}} \hat{\alpha}_r^h.$$

Because the shrinkage factor lies between 0 and 1, the empirical Bayes estimator attenuates each OLS estimate toward the prior mean of zero. Estimates with larger sampling variances (i.e., less precise) are shrunk more heavily, while those estimated with greater precision remain closer to their OLS values.

Imputation of Missing Scores: For a physician i at hospital h from region $r(i)$ who does *not* have an observed licensing score, we impute her score using the empirical Bayes-adjusted model:

$$\hat{y}_{ih} = \hat{\mu}_h + \tilde{\alpha}_{r(i)}^h,$$

where $\hat{\mu}_h$ is the average score of physicians working at hospital h . Thus, the imputed score is a combination of the grand mean for hospital h ($\hat{\mu}_h$), and the “shrunk” region-of-origin differential at hospital h ($\tilde{\alpha}_{r(i)}^h$).

In very few cases (<1%), the data do not include a region of training for the physician. In these cases, we predict the individual score using a LASSO model that incorporates hospital indicators as well as controls for physicians’ age and gender.

B Other Labor Inputs

To work in a public hospital, foreign-trained healthcare professionals must validate their degree in Chile. This process is overseen by specific recognized Chilean universities and involves submitting academic credentials, with additional coursework or exams required in some cases. For physicians, passing the EUNACOM is also necessary, because it qualifies them to practice in public healthcare settings.⁴⁸ After validation, they must register with the National Registry of Individual Health Providers by providing their validated degree and other required personal documentation.

We observe the impact of the migration wave only in the inflow of foreign-trained physicians. As described in the main text, there is a significant increase in the share of foreign-trained physicians, both among those newly enrolled in the National Registry of Healthcare Providers and those working in public hospitals. However, we do not find similar effects for other healthcare workers. As shown in Panel A of Figure A.6, the share of foreign-trained newly registered healthcare workers in the National Registry of Individual Health Providers remained fairly stable between 2000 and 2019, except for physicians. Panel B highlights the share of foreign-trained healthcare workers employed in public hospitals, and shows a significant increase for physicians—from 7% in 2011 to 18% in 2019. In contrast, there is no noticeable change in the share of foreign-trained workers in other healthcare categories.

C Microfoundation for the Aggregate Production Function and the Quantity-Quality Trade-off

C.1 Microfoundation for the Aggregate Production Function

In this appendix, we provide a microfoundation for the aggregate health production function in Equation (5), deriving it from a patient-level model in which physician quantity affects survival directly and through waiting times, while physician quality affects clinical effectiveness and may also affect waiting times by changing the service rate, under random physician-patient matching.

Consider a hospital j at time t with L_{jt} physicians. The outcome of the medical procedure (survival) depends on the patient's health capital at the time of treatment h_{it} , the quality of the treating physician ℓ , $\theta_{\ell jt}$, and the number of physicians available L_{jt} . Here t indexes a period long enough for the queuing system to reach its steady state. Specifically, we consider the patient-level production function for survival probability $y_{i\ell jt}$:

$$y_{i\ell jt} = F(h_{it}, \theta_{\ell jt}, L_{jt}). \quad (\text{A.1})$$

Patient i arriving during period t has initial health capital h_{it}^0 , captured by observable patient

⁴⁸Passing the EUNACOM simultaneously qualifies physicians to practice and validates their degree, which reduces the incentive to pursue a separate degree validation process before taking the EUNACOM.

characteristics X_{it} and an unobservable component u_{it} . Health capital depreciates continuously while the patient waits within the period; given an expected waiting time W_{jt} , health capital at the moment of treatment is:

$$h_{it} = h_{it}^0 - D(W_{jt}). \quad (\text{A.2})$$

The steady state waiting time W_{jt} is determined by the arrival rate of patients as well as the service rate. For simplicity, we introduce the following sufficient conditions in the microfoundation:

Assumption 1 (Stable M/M/L Queuing). *We assume that (i) patients arrive according to a Poisson process with rate λ_{jt} , (ii) service times are exponentially distributed with mean rate $\mu(\bar{\theta}_{jt})$, and (iii) the system is stable, i.e., the traffic intensity $\rho_{jt} \equiv \frac{\lambda_{jt}}{L_{jt}\mu(\bar{\theta}_{jt})} < 1$.*

Assumption 2 (Random Matching). *The specific physician ℓ assigned to patient i is drawn from the within-hospital physician distribution $H_{\theta_{jt}}$, with mean $\bar{\theta}_{jt}$, independent of the patient's initial health capital h_{it}^0 .*

To derive our aggregate production function, we consider the following patient-level (match-specific) survival probability:

$$y_{i\ell jt} = \exp(A_{jt}) \cdot L_{jt}^a \cdot \exp(b\theta_{\ell jt}) \cdot \exp(ch_{it}) \cdot \nu_{i\ell jt}, \quad (\text{A.3})$$

where $\exp(A_{jt})$ captures hospital-specific productivity (TFP) and $\nu_{i\ell jt}$ is an idiosyncratic shock with $\mathbb{E}[\nu] = 1$. Because $y_{i\ell jt}$ is a survival probability, we assume the three structural parameters are positive: $a > 0$ (more physicians improve survival), $b > 0$ (higher physician quality improves survival), and $c > 0$ (greater health capital at the time of treatment improves survival).

We aggregate the previous outcome equation over the distribution of physician quality and patient heterogeneity to obtain the expected hospital outcome y_{jt}^k :

$$y_{jt}^k = \mathbb{E}[y_{i\ell jt}] = \exp(A_{jt}) L_{jt}^a \cdot \mathbb{E}_{\theta}[\exp(b\theta_{\ell jt})] \cdot \mathbb{E}_{h^0}[\exp(c(h_{it}^0 - D(W_{jt})))] . \quad (\text{A.4})$$

Assuming $\theta_{\ell jt} \sim \mathcal{N}(\bar{\theta}_{jt}, \sigma_{\theta}^2)$, the term $\mathbb{E}[\exp(b\theta_{\ell jt})] = \exp(b\bar{\theta}_{jt} + \frac{1}{2}b^2\sigma_{\theta}^2)$. Taking logs of the aggregate outcome y_{jt}^k yields:

$$\ln y_{jt}^k = A_{jt} + a \ln L_{jt} + b\bar{\theta}_{jt} + \text{const} + \ln \mathbb{E}_{h^0}[\exp(ch_{it}^0)] - c D(W_{jt}). \quad (\text{A.5})$$

where we define the patient case-mix term

$$\Gamma_{jt} \equiv \ln \mathbb{E}_{h^0}[\exp(ch_{it}^0)] , \quad (\text{A.6})$$

which is captured empirically by patient risk controls D_{jt} . The remaining term, the depreciation

loss $cD(W_{jt})$, depends on the equilibrium waiting time, which under Assumption 1 is given by:

$$W_{jt} = \frac{C(L_{jt}, \rho_{jt})}{L_{jt}\mu(\bar{\theta}_{jt}) - \lambda_{jt}}, \quad (\text{A.7})$$

where $C(L_{jt}, \rho_{jt})$ is the Erlang-C formula, representing the probability that an arriving patient finds all L_{jt} physicians busy and must wait. The steady state waiting time makes explicit that waiting times are decreasing in the quantity and quality of physicians, where the latter is governed by how much $\mu(\cdot)$ increases as a function of $\bar{\theta}_{jt}$.

Following our previous functional form specifications, we assume $\mu(\bar{\theta}_{jt}) = \mu_0 \exp(\phi \bar{\theta}_{jt})$, and $D(W) = \delta W_{jt}$, implying a constant marginal effect from delay. As a special case, when $\phi = 0$, quality does not affect waiting times.

While the exact structural function $\delta W(\cdot)$ is highly non-linear, we approximate the depreciation loss using a first-order Taylor expansion with respect to inputs $\ln L_{jt}$, $\bar{\theta}_{jt}$, and demand volume $\ln \lambda_{jt}$ around the sample means:

$$\delta W_{jt} \approx d_0 + \underbrace{\delta \frac{\partial W}{\partial \ln L}}_{d_L < 0} \ln L_{jt} + \underbrace{\delta \frac{\partial W}{\partial \bar{\theta}}}_{d_\theta < 0} \bar{\theta}_{jt} + \underbrace{\delta \frac{\partial W}{\partial \ln \lambda}}_{d_\lambda > 0} \ln \lambda_{jt}. \quad (\text{A.8})$$

Each slope d_L, d_θ, d_λ bundles the marginal effect rate δ with the marginal effects of equilibrium waiting time to each input, so that $d_L < 0, d_\theta < 0$, whenever $\phi > 0$, and $d_\lambda > 0$. This approximation incorporates the fact that waiting times are determined not just by supply (L, θ) but also by demand intensity (λ) . Combining this with the aggregate production function yields:

$$\ln y_{jt}^k = (a - c d_L) \ln L_{jt} + (b - c d_\theta) \bar{\theta}_{jt} + \mathbf{O}'_{jt} \boldsymbol{\beta}_O^k + E_{jt}, \quad (\text{A.9})$$

where the composite error term $E_{jt} \equiv \xi_{jt} - c d_\lambda \ln \lambda_{jt}$ captures both productivity shocks and unobserved demand-driven congestion.

In our empirical specification, we absorb the variation in arrival rates $\ln \lambda_{jt}$ using hospital fixed effects (permanent scale), time fixed effects (aggregate demand trends), and time-varying controls (observable case mix changes).

This derivation clarifies that our reduced-form coefficients capture both the direct technical parameters (a, b) and the indirect health channels (via d_L, d_θ). Comparing with Equation (5), the output elasticity with respect to physician quantity is $\alpha_L^k = a - c d_L$, which combines the direct caseload effect (a) with the indirect health effect from reduced waiting $(-c d_L)$; analogously, the output elasticity with respect to physician quality is $\alpha_\theta^k = b - c d_\theta$, combining the direct treatment-quality effect (b) with the indirect effect from faster service $(-c d_\theta)$.

The hospital and time fixed effects, demographics, and other controls in Equation (5) correspond to $\mathbf{O}'_{jt} \boldsymbol{\beta}_O^k$, while the composite error u_{jt}^k absorbs both the unobserved demand-driven congestion

term $-c d_\lambda \ln \lambda_{jt}$ and the measurement error from using scores $\bar{s}_{jt}$ to proxy for latent quality $\bar{\theta}_{jt}$. When the outcome is mortality $m = 1 - p$, where p denotes survival, the log-mortality coefficients equal $-p/(1-p)$ times the log-survival coefficients, so evaluating this factor at the baseline survival rate gives a local constant-coefficient approximation with $\alpha_L^k < 0$ and $\alpha_\theta^k < 0$.

C.2 The Quantity-Quality Trade-off

The aggregate production function derived in Appendix C.1 makes the licensing trade-off explicit. With some abuse of notation, fix a period and consider a hospital with $L - 1$ incumbent physicians of average quality $\bar{\theta}$, and suppose a marginal physician of quality θ_m is added to the hospital's staff, raising the physician count to L . Under Assumption 2, each patient is now equally likely to be treated by any of the L physicians, so the average quality a patient faces becomes

$$\bar{\theta}' = \frac{(L-1)\bar{\theta} + \theta_m}{L}, \quad \text{so that} \quad \bar{\theta}' - \bar{\theta} = \frac{\theta_m - \bar{\theta}}{L}. \quad (\text{A.10})$$

For sufficiently large L , adding the physician raises the physician count by $\Delta \ln L = \ln L - \ln(L-1) \approx 1/L$, and shifts average quality by $\Delta \bar{\theta} = (\theta_m - \bar{\theta})/L$. Applying the log-linearized production function derived above, with output elasticities $\alpha_L^k = a - c d_L$ and $\alpha_\theta^k = b - c d_\theta$, the resulting change in expected survival is, to first order,

$$\Delta \ln y^k \approx \alpha_L^k \Delta \ln L + \alpha_\theta^k \Delta \bar{\theta} = \frac{1}{L} \left[\alpha_L^k + \alpha_\theta^k (\theta_m - \bar{\theta}) \right]. \quad (\text{A.11})$$

These two terms capture the policy's competing forces: $\alpha_L^k > 0$ is a quantity gain experienced by every patient regardless of the entrant's quality, while $\alpha_\theta^k (\theta_m - \bar{\theta})$ is a quality effect, since under random matching a below-average entrant ($\theta_m < \bar{\theta}$) dilutes the quality faced by every patient. The marginal physician therefore raises expected survival if and only if

$$\underbrace{\alpha_L^k}_{\text{quantity gain}} > \underbrace{\alpha_\theta^k (\bar{\theta} - \theta_m)}_{\text{quality loss}}, \quad (\text{A.12})$$

making explicit the quantity-quality trade-off in our setting.

D Do Other Inputs Affect the Results?

In this appendix, we first document the existence of complementarities between physicians and other healthcare workers at the hospital level and then discuss the implications for identifying physicians' returns on hospital outcomes.

Panel A in Appendix Figure A.9 presents a scatter plot of the number of physicians and other full-time equivalent healthcare workers at the hospital. An observation is a hospital-year. The figure

shows a strikingly linear relationship between physicians and other inputs. We further examine whether the mix of other inputs relative to physicians changed with the increase in the number of physicians by differentiating between the pre- and post-migration wave and do not find differences between them. Since both sets of observations overlap, this provides evidence that as hospitals hired more physicians, they hired additional inputs at a similar rate.

A related question is whether the mix of physicians and other healthcare workers varies according to physician quality. In Panel B, we plot the ratio of other healthcare workers to physicians against average physician quality. We do not find that hospitals with higher-quality physicians substitute away from other inputs.

Given this evidence, it is natural to ask about the implications of incorporating other inputs in the production function as a function of the number of physicians. Since the ratio of other inputs to physicians does not change with physician quality, we assume away quality for ease of exposition. The production function takes the following form: $Y_i = AL_i^{\alpha_L} O_i^{\alpha_O}$, with $O_i = e^c L_i^\gamma$. Thus,

$$\ln Y_i = \phi + \underbrace{(\alpha_L + \gamma\alpha_O)}_{\tilde{\alpha}_L} \ln L_i.$$

Given the complementarities between physicians and other inputs, the impact of physicians on the outcome of interest, $\tilde{\alpha}_L$, is a bundled effect: a direct effect of an extra doctor, α_L , and an indirect effect from the increase in other inputs, $\gamma\alpha_O$.

We next discuss the implications for identification in our setting, in which the quantity of physicians and the number of other inputs are endogenous. Assume that we specify the following estimating equation:

$$y_i = \phi + \alpha_L l_i + \alpha_O o_i + \epsilon_i,$$

where y_i is a logged outcome; l_i is the logged quantity of doctors; and o_i is the logged number of other healthcare workers. Assume that l_i and o_i are endogenous (i.e., $\text{Cov}(l_i, \epsilon_i) \neq 0$, and $\text{Cov}(o_i, \epsilon_i) \neq 0$). Further, assume a plausibly exogenous instrument z_i for l_i , such that $\text{Cov}(l_i, z_i) \neq 0$, but which might be correlated with o_i ($\text{Cov}(z_i, o_i) \neq 0$), which could violate the exclusion restriction. The IV estimator of α_L is given by

$$\begin{aligned} \hat{\alpha}_L^{IV} &= \frac{\text{Cov}(y_i, z_i)}{\text{Cov}(l_i, z_i)} \\ &= \alpha_L + \alpha_O \frac{\text{Cov}(o_i, z_i)}{\text{Cov}(l_i, z_i)}, \end{aligned}$$

where the second step follows from the assumption that $\text{Cov}(\epsilon_i, z_i) = 0$. Incorporating the comple-

mentarity between physicians and other inputs as $o_i = c + \gamma l_i + \nu_i$,

$$\begin{aligned}\hat{\alpha}_L^{IV} &= \alpha_L + \alpha_O \frac{\text{Cov}(c + \gamma l_i + \nu_i, z_i)}{\text{Cov}(l_i, z_i)} \\ &= \underbrace{\alpha_L + \gamma \alpha_O}_{\tilde{\alpha}_L} + \underbrace{\alpha_O \frac{\text{Cov}(\nu_i, z_i)}{\text{Cov}(l_i, z_i)}}_{\text{bias}}.\end{aligned}$$

Thus, $\hat{\alpha}_L^{IV}$ is the bundled effect of the direct and indirect effects of more physicians on the outcome, plus a bias term if $\text{Cov}(\nu_i, z_i) \neq 0$ (i.e., if innovations that shift o_i correlate with z_i). We can empirically gauge the magnitude of the bias by estimating the linear projection of the instrument on other inputs via OLS:

$$o_i = \alpha + \beta z_i + e_i.$$

Using the fact that $\beta = \frac{\text{Cov}(o_i, z_i)}{\text{Var}(z_i)}$, it follows that:

$$\hat{\text{Cov}}(\nu_i, z_i) = \hat{\beta} \hat{\text{Var}}(z_i) - \gamma \hat{\text{Cov}}(l_i, z_i).$$

Empirically, we obtain $\hat{\beta} = 0.01$, $\hat{\text{Var}}(z_i) = 0.86$, $\hat{\text{Cov}}(l_i, z_i) = 0.02$, and $\hat{\gamma} = 0.41$.⁴⁹ Adding these numbers, we find that $\hat{\text{Cov}}(\nu_i, z_i) = 0.0006 \approx 0$, which we interpret as evidence of negligible bias. Complementary evidence for these results comes from the trends in labor supply of L_i and O_i over time, as discussed in Appendix B. The evidence therein shows that migration shocks mostly shifted the supply of physicians, with negligible effects on other inputs.

E Quality Gains

In general, our model could allow for scores to improve over attempts due to increased test-taking ability as well as improvements in quality. With data on scores alone, it is not possible to disentangle both mechanisms. In this section, we show that auxiliary data are consistent with no quality improvements over attempts. This justifies the assumption of no quality gains over attempts.

We leverage the discontinuity in retaking probability around the passing threshold to show that retaking does not improve labor-market outcomes that proxy for quality (Fleitas, 2020). Figure A.7-A shows the probability of ever retaking the test as a function of the distance from the score on the first attempt to the threshold. It shows that the retaking probability drops substantially around the cutoff. Figure A.7-B shows that the maximum achieved score changes discontinuously around the threshold. Note that score gains in Panel B could be explained by a combination of

⁴⁹Note that, for this exercise, $\hat{\gamma}$ is estimated from an instrumental-variables regression of o_i on l_i , using the second lag of l_i as an instrument. Using this alternative instrument rather than z_i ensures that the estimate of γ is not mechanically related to the object of interest, $\widehat{\text{Cov}}(\nu_i, z_i)$.

gains in test-taking ability, gains in quality, and selection around the cutoff (Gilraine and Penney, 2025).

We investigate how two proxies of quality change around the threshold. Figure A.7-C shows the probability of being selected as *Médico General de Zona*—a merit-based position for physicians in Chile who serve in underserved areas and grants eligibility for state-funded specialization. The likelihood increases substantially with the score on the first attempt. However, there are no discernible increases around the threshold. Similarly, Figure A.7-D shows a (more modest) increase in contract with the score on the first attempt, with no discernible increases around the threshold. Together, we take this as evidence that contradicts the idea that there are quality improvements from retaking.

F Microfoundation of CMPs

We now provide a microfoundation for the reduced-form CMPs presented in the main text. We consider a static version of the model and fix t . Consider a continuum of physicians indexed by $i \in \mathcal{I}$. Each physician i belongs to a type $\tau_i \in \{N, F\}$, with a total mass m_τ for each type τ . Let $\mathcal{J}$ be a finite set of hospitals, each indexed by j and endowed with $\kappa_j \in \mathcal{K}$ vacancies, and let ϕ be physicians’ outside option.

Physician preferences. The physician’s indirect utility from matching with hospital $j \in \mathcal{J}$ is given by

$$u_{ij} = \tilde{v}(x_{ij}, \hat{\theta}_i) + \varepsilon_{ij}, \quad (\text{A.13})$$

Hospital preferences. Hospitals maximize an objective function that is strictly increasing in the expected quality $\hat{\theta}$ of the physicians they match with and find all to be admissible. Strict monotonicity in $\hat{\theta}$ implies that if a hospital j can replace a physician of lower expected quality with one of higher expected quality, it will strictly prefer to do so.

Equilibrium. An equilibrium allocation satisfies *pairwise stability*:

- (i) *Individual rationality*: No matched physician prefers their outside option to their allocation.
- (ii) *No blocking pairs*: No hospital-physician pair wants to deviate from their current match and match with each other.

Following Azevedo and Leshno (2016), the equilibrium can be characterized by a system of demand and supply equations that determine a set of—expected quality—cutoffs. In particular, each hospital j admits any physician i with $\hat{\theta}_i$ above their cutoff $\hat{\theta}_j$, while each physician i chooses the hospital

j that maximizes her indirect utility from among those that would admit her. Mathematically, i matches with j —i.e., $\mu(i) = j$, if and only if

$$\mu(i) = \arg \max_{j \in \mathcal{J}_i} u_{ij},$$

where $\mathcal{J}_i = \{j \in \mathcal{J} : \hat{\theta}_i \geq \hat{\theta}_j\} \cup \{\phi\}$.

Given our distributional assumptions on physicians' unobserved preferences, the conditional matching probability for a physician i matching with hospital j can be written as

$$CMP^m(\mu(i) = j) = \frac{e^{\tilde{v}(x_{ij}, \hat{\theta}_i)} \mathbf{1}\{\hat{\theta}_i \geq \hat{\theta}_j\}}{1 + \sum_{j'} e^{\tilde{v}(x_{ij'}, \hat{\theta}_i)} \mathbf{1}\{\hat{\theta}_i \geq \hat{\theta}_{j'}\}}, \quad (\text{A.14})$$

where the indicator $\mathbf{1}\{\hat{\theta}_i \geq \hat{\theta}_j\}$ requires that we only consider hospitals that are ex post feasible for physician i (Fack et al., 2019), and the equilibrium cutoffs are determined by the following system of demand and supply equations:

$$\kappa_j \geq M_j(\hat{\underline{\theta}}) \forall j \in \mathcal{J}, \quad (\text{A.15})$$

with equality whenever hospital j 's quality cutoff is binding, where

$$M_j(\hat{\underline{\theta}}) = \sum_{\tau} m_{\tau} \int_{\hat{\underline{\theta}}_j}^{\hat{\bar{\theta}}} \int_{\mathcal{X}} \frac{e^{\tilde{v}(X, y|j)} \mathbf{1}\{y \geq \hat{\theta}_j\}}{1 + \sum_{j'} e^{\tilde{v}(X, y|j')} \mathbf{1}\{y \geq \hat{\theta}_{j'}\}} h(X, y|\tau) dX dy \quad (\text{A.16})$$

with $\hat{\underline{\theta}}$ denoting the vector of posterior quality cutoffs, $h(X, y|\tau)$ the joint density over observable characteristics X and posterior quality y given type τ , and $\hat{\bar{\theta}}$ the upper bound in the domain of posterior mean qualities.

The previous equations make explicit the dependency of equilibrium cutoffs on hospitals' vacancies and the mass and joint distribution of physicians' preferences and posterior qualities.

Approximation of CMPs. We now provide conditions under which our reduced-form CMPs can be seen as approximations of the microfounded CMPs. The reduced-form CMPs for this simplified model, specified in Equation (8), would be given by

$$CMP^r(\mu(i) = j) = \frac{e^{v(x_{ij}, \hat{\theta}_i) + g(\mathbf{M}_i(\underline{s}), \kappa|j)}}{1 + \sum_{j'} e^{v(x_{ij'}, \hat{\theta}_i) + g(\mathbf{M}_i(\underline{s}), \kappa|j')}}. \quad (\text{A.17})$$

Therefore, the reduced-form CMPs given in Equation (A.17) can approximate the microfounded

CMPs from Equation (A.14) if:

$$e^{\Delta_v(x_{ij}, \hat{\theta}_i) + g(\mathbf{M}_i(\underline{s}), \kappa|j)} \approx \mathbf{1}\{\hat{\theta}_i \geq \hat{\theta}_j\} \quad \forall j \in \mathcal{J}, \quad (\text{A.18})$$

where $\Delta_v(x_{ij}, \hat{\theta}_i) \equiv v(x_{ij}, \hat{\theta}_i) - \tilde{v}(x_{ij}, \hat{\theta}_i)$.⁵⁰

To establish under which conditions Equation (A.18) might hold, notice that the indicator function can be smoothed out by using the Gompertz function,

$$e^{-e^{-\lambda(\hat{\theta}_i - \hat{\theta}_j)}} \approx \mathbf{1}\{\hat{\theta}_i \geq \hat{\theta}_j\} \quad \forall j \in \mathcal{J},$$

which converges to an indicator as $\lambda \rightarrow \infty$. Thus, our reduced-form CMPs can approximate their microfounded counterparts if

$$\Delta_v(x_{ij}, \hat{\theta}_i) + g(\mathbf{M}_i(\underline{s}), \kappa|j) \approx -e^{-\lambda(\hat{\theta}_i - \hat{\theta}_j)}. \quad (\text{A.19})$$

Equation (A.19) suggests that for our reduced-form CMPs to be a reasonable approximation of their microfounded counterparts, the function $g(\cdot)$ must capture the determinants of the equilibrium cutoffs $\hat{\theta}$. As shown in Equations (A.15) and (A.16), these equilibrium cutoffs are the solutions to a system of supply and demand equations that depend on the distribution of preferences, the masses of physicians, and hospitals' capacities.

In counterfactual analyses, changes in the mass of physicians with different preferences and qualities might affect the equilibrium cutoffs in complex ways. However, under the assumption of vertical sorting, an important restriction arises: the (ex post feasible) choice set—and thus the matching probabilities—of a physician with posterior quality $\hat{\theta}_i$ can only be affected by changes in the mass of physicians with qualities above $\hat{\theta}_i$.

To see this more concretely, consider a large matching market with a finite set of agents and a unique stable matching. In this case, the equilibrium allocation coincides with the outcome of the serial dictatorship (SD) mechanism (Abdulkadiroğlu and Sönmez, 1998), in which physicians are ordered by descending posterior quality and sequentially choose their most preferred hospital among those with remaining vacancies. Since hospitals have vertical preferences, a physician's assignment is determined solely by her relative position in the quality ranking. When physician i makes her choice, the set of available hospitals depends only on the prior selections made by physicians with higher quality. Thus, changes in the mass of physicians with quality below $\hat{\theta}_i$ do not alter the set of options available to her, and consequently leave her allocation and conditional matching probability unchanged.

Motivated by this result from our microfounded model, we specify the function $g(\cdot)$ to depend on

⁵⁰Notice that because $v(x_{ij}, \hat{\theta}_i) = \Delta_v(x_{ij}, \hat{\theta}_i) + \tilde{v}(x_{ij}, \hat{\theta}_i)$, in our reduced-form CMPs we will not be able to separately identify how much of the determinants of matching probabilities (v) are affected by physicians' preferences ($\tilde{v}$), versus the effect of their observable characteristics on their choice sets (Δ_v).

hospitals' vacancies κ and on the mass of physicians with quality above i 's posterior quality, denoted by M_i . In the empirical implementation, we discretize quality into quality ranges for estimation, and thus allow $g(\cdot)$ to depend on both M_{it}^0 (the mass in i 's quality range) and M_{it}^+ (the mass above i 's quality range).

G Posterior Quality in the Dynamic Model of Scores

The latent-quality distribution described in Section 6 is the two-component normal mixture

$$\text{MixN}_2(\tau_i, t_i^0) \equiv \pi_{\tau_i} N(\mu_{\theta, \tau_i, t_i^0, 1}, \sigma_{\theta, \tau_i, t_i^0, 1}^2) + (1 - \pi_{\tau_i}) N(\mu_{\theta, \tau_i, 2}, \sigma_{\theta, \tau_i, 2}^2).$$

Thus, the first component depends on type and entry cohort, whereas the second depends only on type.⁵¹

The standard deviations of component-1 latent quality and of the score noise are parameterized in logs with cohort trends: piecewise-linear for the former (with a 2016 break and a 2015 deviation for nationals) and linear for the latter. The mixture weights, component-2 mean and standard deviation, and (γ, ρ) are time-invariant, with (γ, ρ) common across types.

We use the physician's score history to update this prior distribution and infer posterior quality. For a physician with score history $\vec{s}_i \equiv \{s_{i1}, s_{i2}, \dots, s_{iN_i}\}$ and N_i attempts, define the average attempt-adjusted score as $\bar{y}_i \equiv \frac{1}{N_i} \sum_{k=1}^{N_i} (s_{ik} - \Gamma_{ik})$. The full posterior quality is given by

$$\hat{\theta}(\vec{s}_i | \tau_i, t_i^0) \equiv \mathbb{E}[\theta_i | s_{i1}, \dots, s_{iN_i}, \tau_i, t_i^0] = \sum_{m \in \{1, 2\}} \omega_{im} \left(\mu_m + \frac{N_i \sigma_m^2}{\sigma_{\epsilon, \tau_i, t_i^0}^2 + N_i \sigma_m^2} (\bar{y}_i - \mu_m) \right). \quad (7)$$

Here $(\mu_1, \sigma_1^2) = (\mu_{\theta, \tau_i, t_i^0, 1}, \sigma_{\theta, \tau_i, t_i^0, 1}^2)$ and $(\mu_2, \sigma_2^2) = (\mu_{\theta, \tau_i, 2}, \sigma_{\theta, \tau_i, 2}^2)$ are the component means and variances, while $\pi_1 = \pi_{\tau_i}$ and $\pi_2 = 1 - \pi_1$ are their prior probabilities. The posterior responsibility on component m is $\omega_{im} = \tilde{\omega}_{im} / \sum_{m' \in \{1, 2\}} \tilde{\omega}_{im'}$, where $\tilde{\omega}_{im} = \pi_m \phi(\bar{y}_i; \mu_m, \sigma_m^2 + \sigma_{\epsilon, \tau_i, t_i^0}^2 / N_i)$ and $\phi(\cdot; \mu, \sigma^2)$ denotes the normal density with mean μ and variance σ^2 .

Equation (7) is the basis for inferring a physician's posterior quality from her score history and the estimated parameters of the dynamic model of scores. Intuitively, within each mixture component, the posterior mean is a weighted average of its prior mean and the physician's attempt-adjusted average score, with more weight on the score when the signal is more informative—that is, when the physician has more attempts or score noise is lower. This weight is the multi-attempt analog of $\sigma_\theta^2 / (\sigma_\theta^2 + \sigma_\epsilon^2)$. The final posterior averages these component-specific means using their posterior responsibilities. The parameters (γ, ρ) enter the posterior through the attempt adjustments in $\bar{y}_i$, while the score-noise variance and mixture parameters determine the within-component weights

⁵¹The two-component mixture captures the empirical distribution of scores well (see Appendix Figure A.3). We allow only the dominant component to vary by cohort, since the second's small mass limits identification of cohort-specific parameters.

and posterior responsibilities.

H Definition and Computation of Elasticities

Hospital-specific Elasticities The labor-matching probabilities and the licensing threshold determine the hospital-specific inflows of physicians and their respective quality. For a physician $i \in \mathcal{I}$ with type $\tau \in \mathcal{T}$, exam score s , and a matrix of physician–hospital characteristics $X \in \mathbb{R}^{K \times |\mathcal{J}|}$ (whose j th column is the vector $\mathbf{x}_{ijt} \in \mathbb{R}^K$ for hospital $j \in \mathcal{J}$), in a market characterized by (M_t, κ_t) in period t , we define the conditional probability that such a physician matches with hospital $j \in \mathcal{J}$ in period t as

$$CMP_j(X, s \mid \tau; M_t; \kappa_t; \underline{s}) \equiv CMP\left([x_{i1t}, \dots, x_{i|\mathcal{J}|t}] = X, s_i = s, \tau_i = \tau, M_{it} = M_t, \kappa_t, \underline{s}, j\right).$$

Letting $h_t^\tau(s, X)$ be the type- and time-specific density of the joint distribution of observables and scores, the labor inflow of type τ in hospital j in period t is given by

$$\Delta L_{jt}^\tau(\underline{s}) = m_{\tau,t} \int_X \int_{s \geq \underline{s}} CMP_j(X, s \mid \tau; M_t; \kappa_t; \underline{s}) h_t^\tau(s, X) ds dX.$$

In addition, the average quality of incoming physicians of type τ in hospital j at time t is given by

$$\Delta \bar{\theta}_{jt}^\tau(\underline{s}) = \frac{\int_X \int_{s \geq \underline{s}} \mathbb{E} \left[CMP_{ijt} \hat{\theta}_i \mid s, X \right] h_t^\tau(s, X) ds dX}{\int_X \int_{s \geq \underline{s}} \mathbb{E} [CMP_{ijt} \mid s, X] h_t^\tau(s, X) ds dX},$$

where the conditional expectations average over score histories $\vec{s}$ consistent with s and X , evaluating each physician’s matching probability and posterior quality jointly.

The labor in hospital j at time t and its corresponding average quality are given by

$$L_{jt}(\underline{s}) = L_{j,t-1} + \sum_{\tau \in \mathcal{T}} \Delta L_{jt}^\tau(\underline{s}) \tag{A.20}$$

$$\bar{\theta}_{jt}(\underline{s}) = \frac{1}{L_{jt}} \left(\bar{\theta}_{j,t-1} L_{j,t-1} + \sum_{\tau \in \mathcal{T}} \Delta L_{jt}^\tau(\underline{s}) \cdot \Delta \bar{\theta}_{jt}^\tau(\underline{s}) \right) \tag{A.21}$$

The elasticity of quantity and semi-elasticity of quality with respect to the licensing threshold follow directly from the expressions above.

Computation of Numerical Elasticities To compute the elasticity of labor and quality, we simulate $R = 100$ draws for the EV1 shocks in the estimated CMPs and compute the resulting matches for physicians who take the test in year t and have a score above the corresponding

threshold. Let $\Delta^r L_{jt}$ be the number of newly matched physicians with average quality $\Delta^r \theta_{jt}$ in simulation r . We compute the resulting quantity and quality of each hospital by adding the marginal physicians who match with hospital j in year t to the stock of physicians of hospital j in year $t - 1$. That is, the resulting simulated number of physicians is $L_{jt}^r(\underline{s}) = L_{jt-1} + \Delta^r L_{jt}(\underline{s})$ and the resulting simulated average quality is $\bar{\theta}_{jt}^r(\underline{s}) = \frac{(\bar{\theta}_{jt-1} L_{jt-1} + \Delta^r L_{jt} \Delta^r \theta_{jt}(\underline{s}))}{L_{jt}^r(\underline{s})}$.

We compute the quantity elasticity numerically as

$$\eta_{L_{jt}, \underline{s}}^r \approx \frac{\left(L_{jt}^r(\underline{s} + \Delta \underline{s}) - L_{jt}^r(\underline{s}) \right)}{\Delta \underline{s}} \cdot \frac{\underline{s}}{L_{jt}^r(\underline{s})}$$

and the quality semi-elasticity as

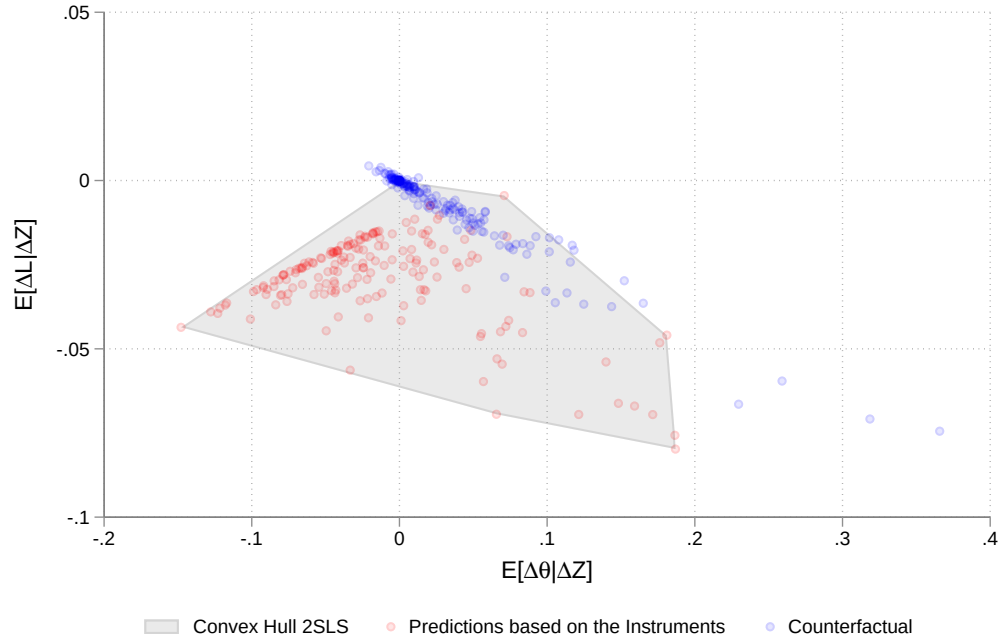
$$\tilde{\eta}_{\bar{\theta}_{jt}, \underline{s}}^r \approx \frac{\left(\bar{\theta}_{jt}^r(\underline{s} + \Delta \underline{s}) - \bar{\theta}_{jt}^r(\underline{s}) \right)}{\Delta \underline{s}} \cdot \underline{s}.$$

We then compute the estimated elasticities by averaging across simulations.

I Relationship between Counterfactual and Identifying Variation

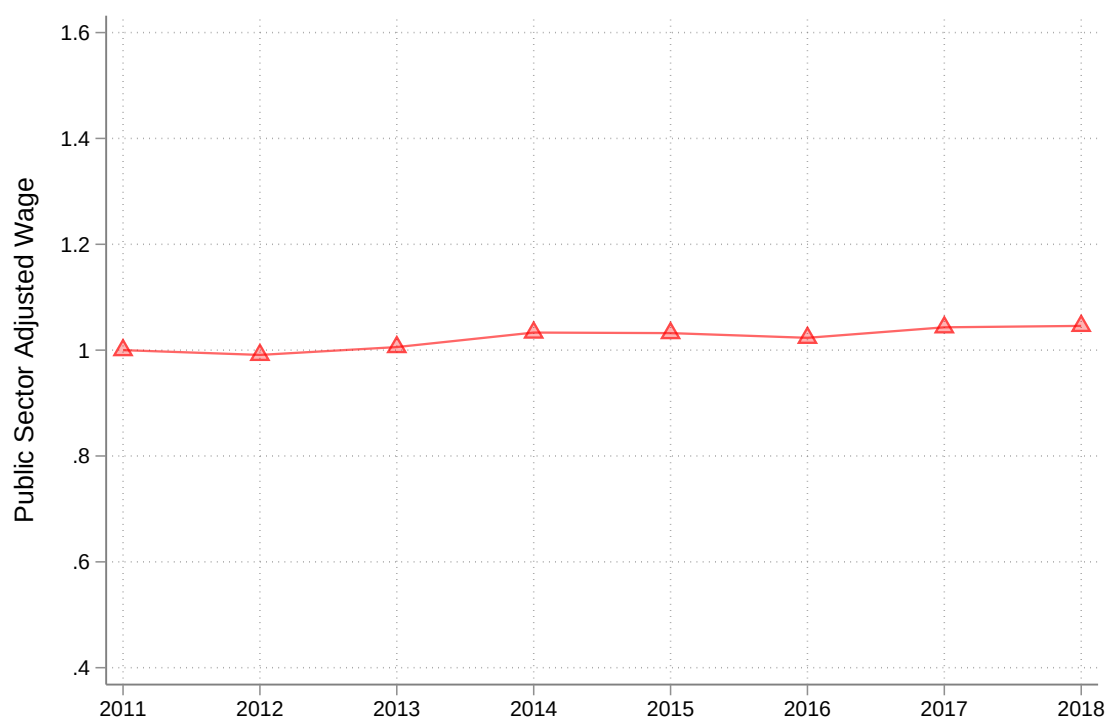
The variation in the labor supply that we exploit is well suited for our counterfactual analysis. As shown in our reduced-form exercise of Section 5.3, the variation in the labor supply experienced in Chile led to a decrease in quality and an increase in logged quantity at an average ratio of 3.2. Our policy counterfactual is qualitatively equivalent to “reversing” the migration wave—increasing the threshold decreases quantity but increases quality. It is also quantitatively similar, in that it increases quality and decreases logged quantity at an average ratio of 4.3. To visually assess the overlap between the changes generated in the counterfactual exercise and the identifying variation caused by the migration wave, Figure A.1 overlays the simulated changes in physician quantity and quality from our counterfactual with the instrument-predicted changes in quantity and quality for each hospital. Reassuringly, the changes induced in the counterfactual analysis are mostly within the convex hull of the identifying variation observed in the data, strengthening our confidence in the robustness of our results to potential misspecification of the production function.

Figure A.1: Convex Hull



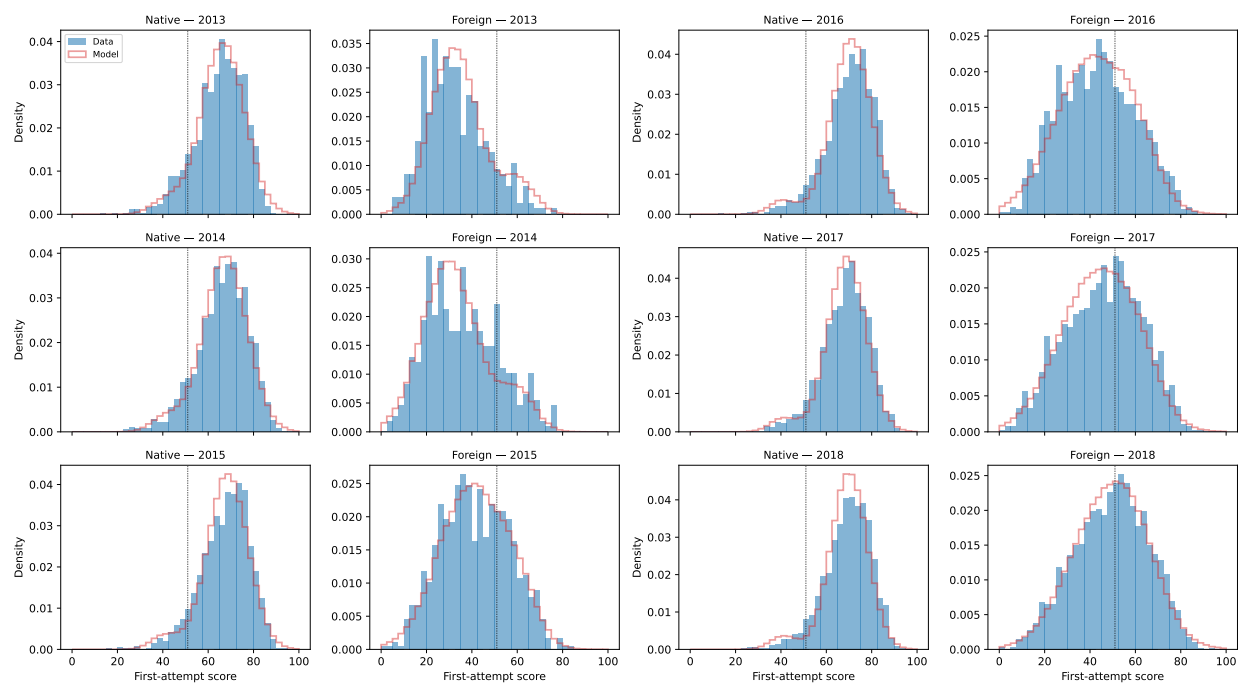
Notes: This figure plots the differences between negative of the instrument-predicted quantity and quality of physicians (in red) and the simulated changes in quantity and quality in our counterfactual (in blue). Predictions are constructed using the first-stage coefficients and average instrument values by hospital. Simulated changes are obtained by rescaling the input elasticities by a factor of 5/51. The convex hull of the instrument's predictions is shown in gray.

Figure A.2: Physicians' Wages in Public Hospitals



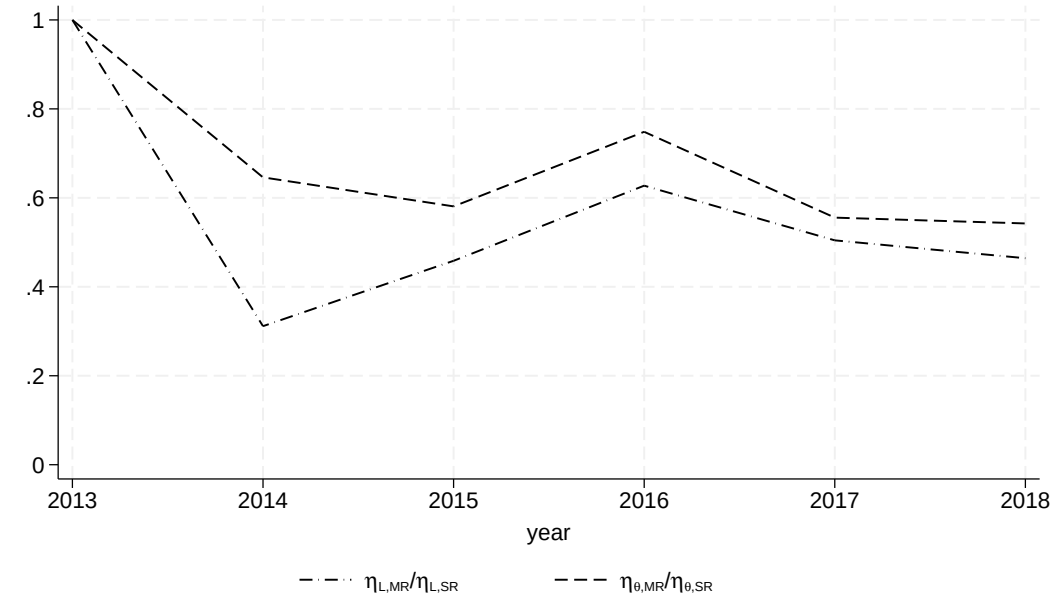
Notes: This figure illustrates the growth of physicians' wages in public hospitals. Wages are adjusted using public-sector-wide adjustment and indexed to 2011, which is set to 1. Data come from [SIRH \(2019\)](#).

Figure A.3: Score Distribution: Data vs. Model, by Type and Cohort

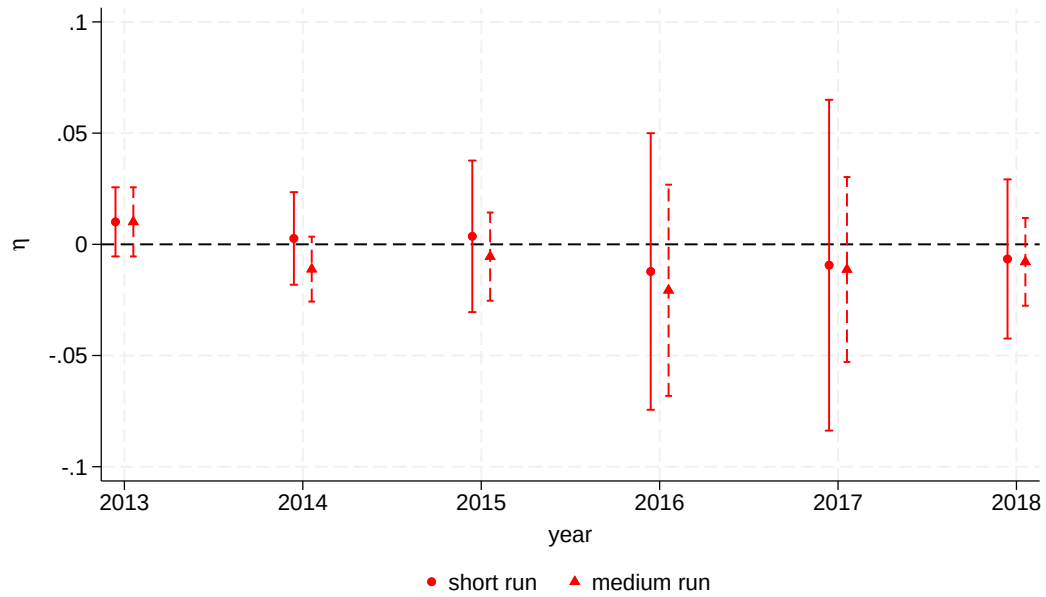


Notes: This figure compares the empirical distribution of first-attempt scores with the distribution implied by the estimated score model, separately by physician type and entry cohort. The blue bars correspond to the data and the red lines correspond to the model-implied distribution. The vertical line represents the passing threshold.

Figure A.4: Simulated Medium- and Short-Run Elasticities



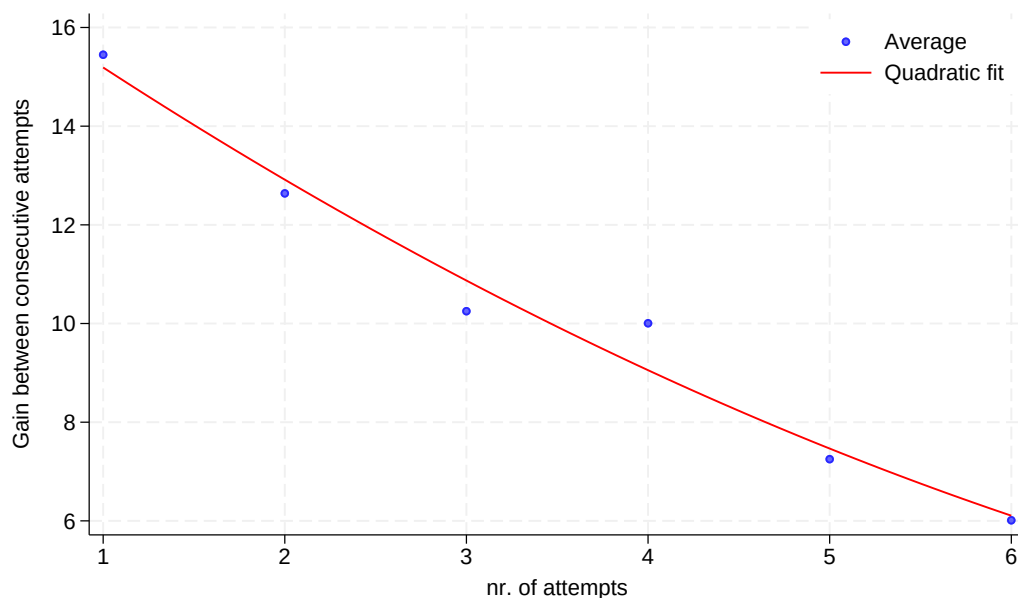
A. Labor Quantity and Quality



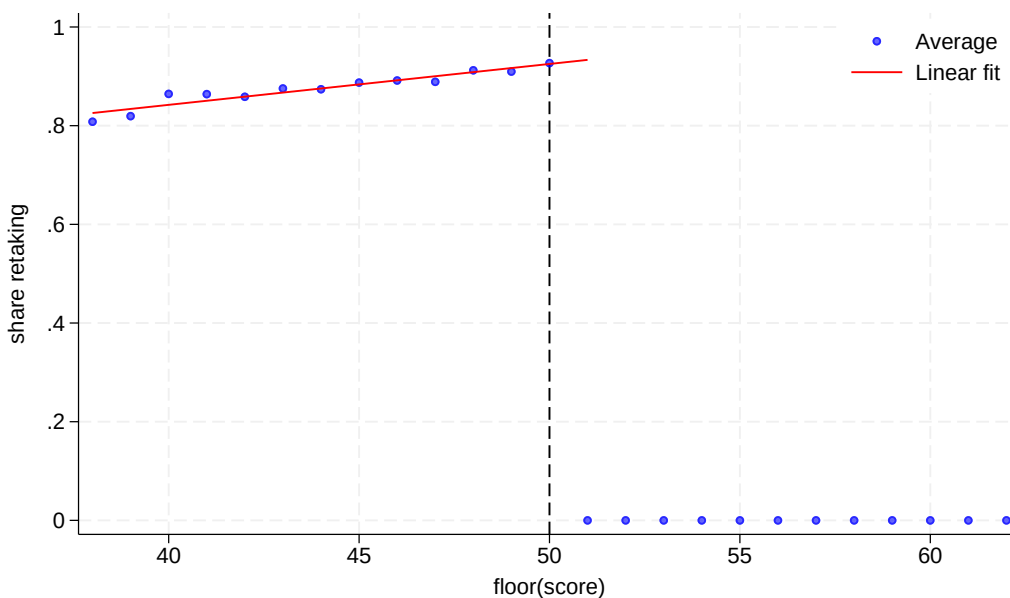
B. Mortality

Notes: Panel A shows the ratio of the simulated medium-run elasticity to the simulated short-run elasticity for physician quantity and quality. Panel B shows the simulated short- and medium-run average effects on patient mortality; the bars are 95% confidence intervals obtained by the delta method from the exposure-robust standard errors of the production-function estimates in Table I. The two elasticities are identical in 2013 by construction.

Figure A.5: Score Gains and Retaking Probability



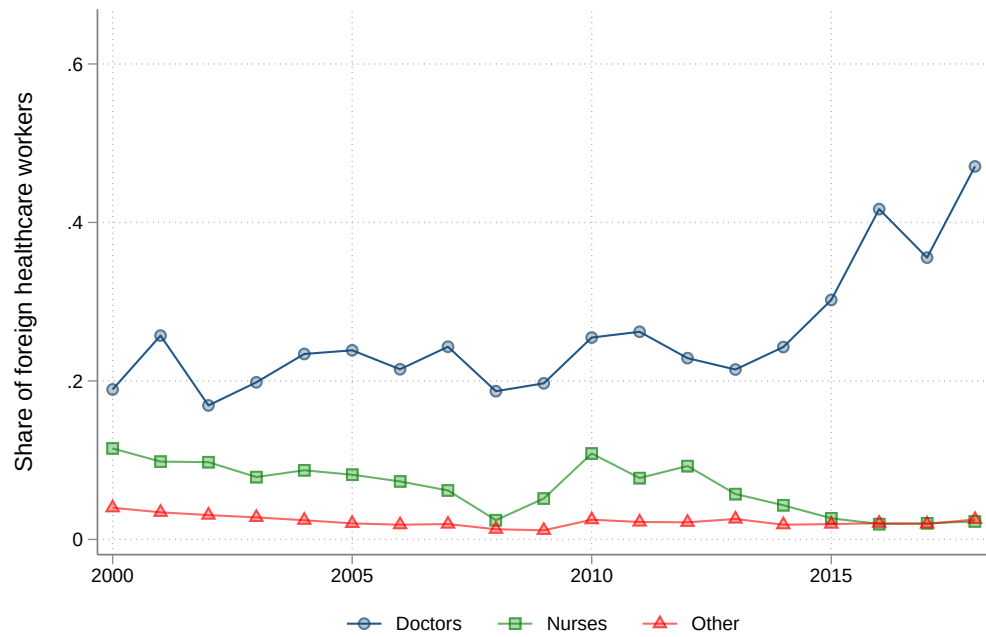
A. Score Gains between Attempts



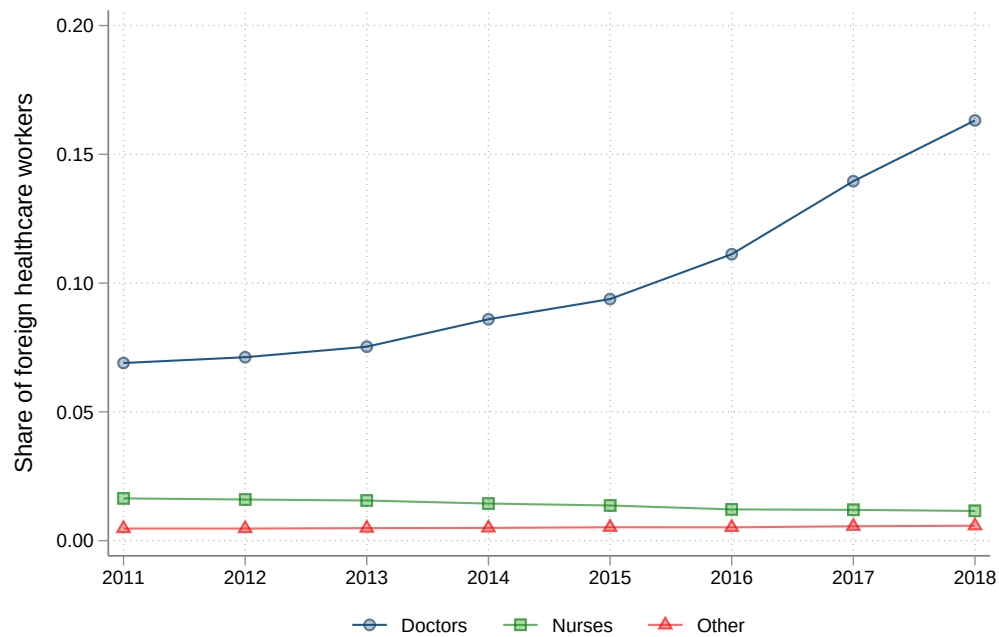
B. Retaking Probability

Notes: Panel A presents the score gains after consecutive test-taking attempts by the same individual. It shows that on average, score gains diminish with each additional attempt and that gains are approximately convex. Data are from [ASOFAMECH \(2019\)](#) and the figure includes the universe of test-takers between 2013 and 2019. Panel B shows the share of test-takers that retake the test as a function of the score in the last attempt for scores between 38 and 62. Among those who fail, most retake, with a retaking probability that increases with the last score. Most test-takers who pass do not retake. The fitted line uses failing scores between 38 and 50.

Figure A.6: Migration Wave: Other Labor Inputs



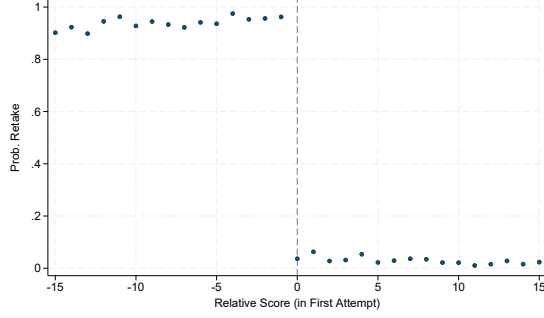
A. Newly Registered Providers



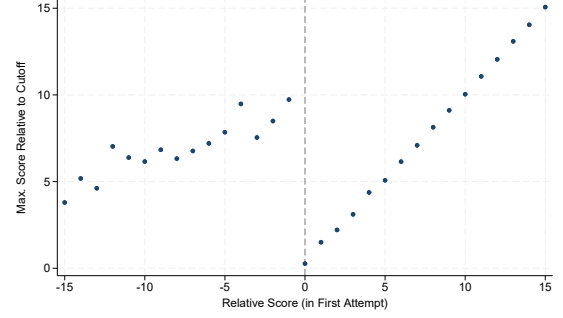
B. Stock in Public Hospitals

Notes: This figure shows the evolution of the share of foreign-trained healthcare providers in Chile. Panel A examines trends from 2000 to 2018 using data from the National Registry of Healthcare Providers (RNPI, 2024). Panel B focuses on healthcare providers working in public hospitals from 2011 to 2018, based on data from SIRH (2019).

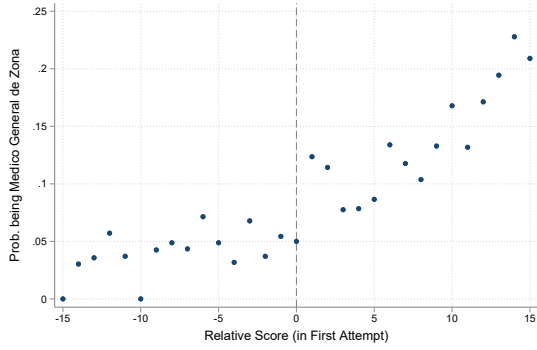
Figure A.7: Testing for Quality Gains



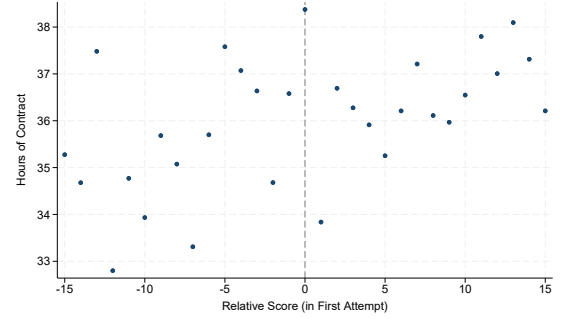
A. Retaking probability



B. Maximum achieved score



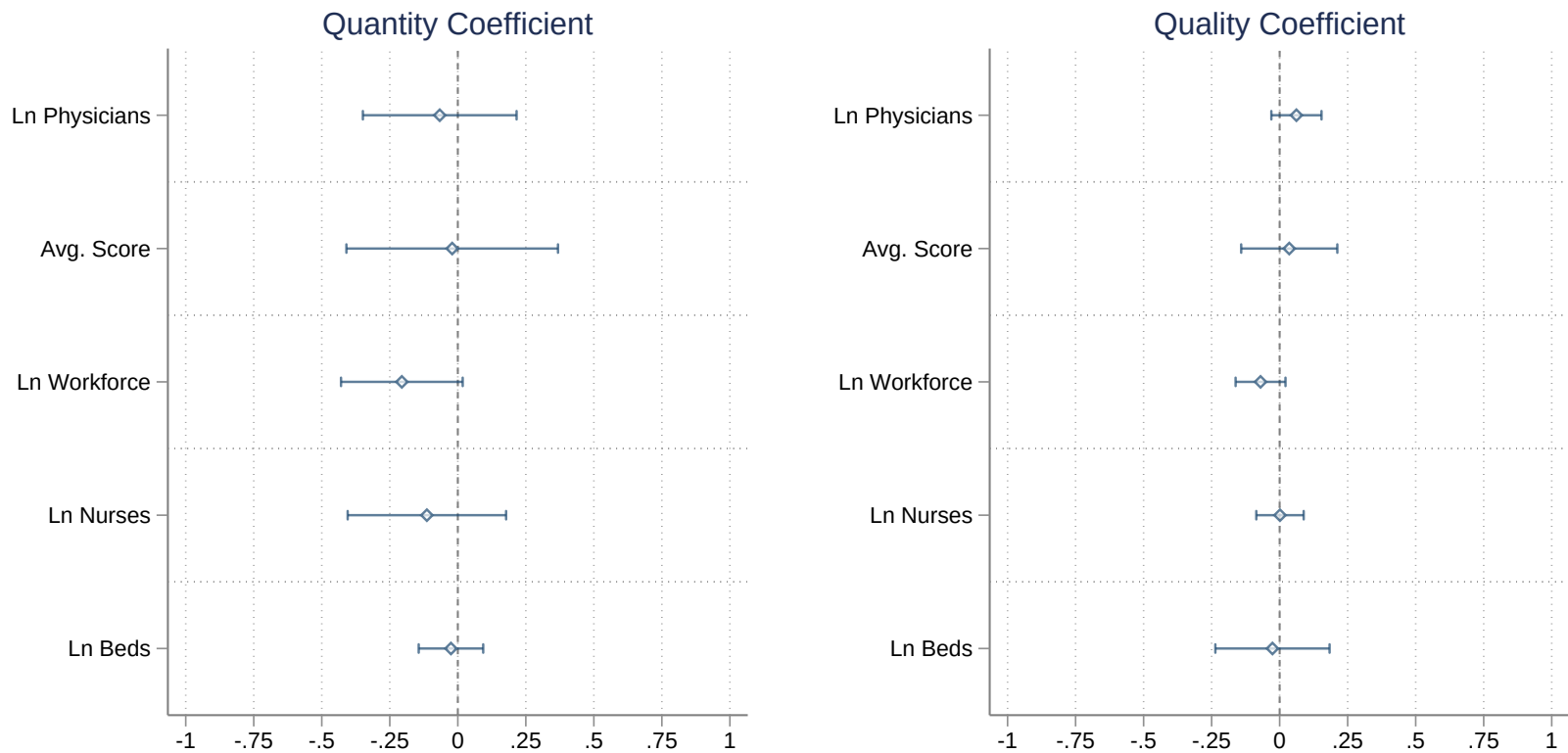
C. Medico General de Zona



D. Hours of contract

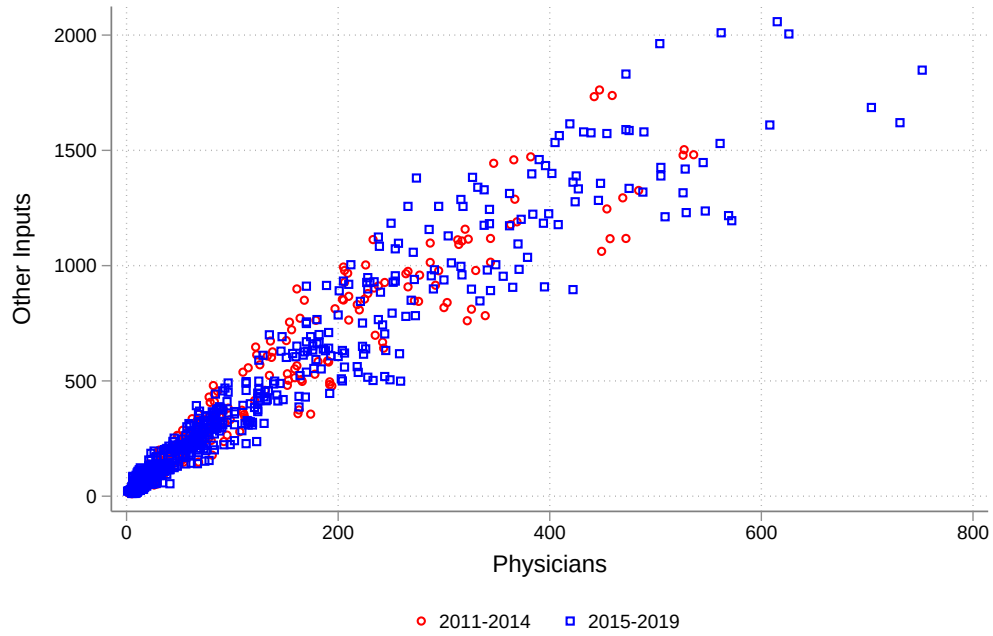
Notes: This figure illustrates the effect of exam retaking on physician-quality improvements. Panel A depicts the probability of ever retaking the exam based on whether they scored above the passing threshold on their first attempt. Panel B presents each individual's maximum achieved score relative to the passing threshold as a function of the first-attempt score relative to the threshold. The line on the right closely follows a 45-degree trajectory, consistent with the fact that individuals who pass the exam do not retake it. In contrast, the line on the left indicates that individuals just below the threshold improve their scores by approximately 10 points on average, with smaller improvements observed for those further below the threshold. Panel C examines the probability of becoming *Medico General de Zona* as a proxy for physician quality (see main text for the definition). There is no visible discontinuity at the threshold on their first test-taking attempt, which is consistent with the absence of quality gains from repeated test-taking. Panel D uses weekly working hours as an additional quality measure, and similarly, there is no visible effect at the threshold. Data on scores are from [ASOFAMECH \(2019\)](#). Panels A and B use the full sample of test-takers, while Panels C and D incorporate data from [SIRH \(2019\)](#), restricting the sample to physicians observed working in public hospitals.

Figure A.8: Shocks Balance Test

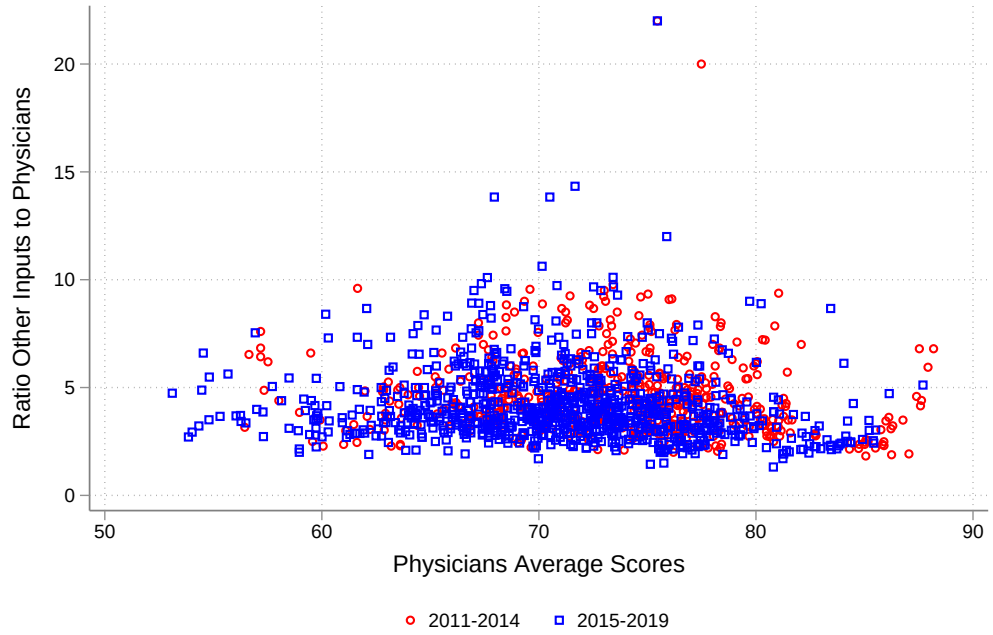


Notes: These figures report coefficients from transformed IV regressions following [Borusyak et al. \(2022\)](#), where changes in hospital-level covariates between 2012 and 2013 are regressed on the shift-share shocks, using only post-2013 variation in the shock series. Regressions are weighted by the hospital-level exposure shares that underlie the instrument and include year fixed effects. Confidence intervals are constructed using exposure-robust standard errors clustered at the region-of-origin level. All variables and shocks are standardized to have mean zero and unit variance.

Figure A.9: Other Hospital Healthcare Workers



A. Other inputs and physicians



B. Mix by physician quality

Notes: This figure examines the relationship between physicians and other full-time equivalent (FTE) healthcare workers in public hospitals. Panel A presents a scatter plot of the number of physicians and other healthcare workers at the hospital level, with each observation representing a hospital-year. Panel B plots the ratio of other healthcare workers to physicians against average physician scores. Pre-migration wave observations are represented by red circle markers, and post-migration wave observations by blue square markers in both panels.

Table A.1: Descriptive Statistics

Panel A: Health Outcomes				
	Mean	Std. Dev.	Median (p50)	# of Obs.
	(1)	(2)	(3)	(4)
Patient Characteristics:				
% Female	0.57	0.07	0.58	1,404
% Foreign	0.01	0.03	0.00	1,404
% Age < 29	0.30	0.15	0.31	1,404
% Age ∈ (30,39)	0.10	0.04	0.10	1,404
% Age ∈ (40,49)	0.09	0.02	0.09	1,404
% Age ∈ (50,59)	0.11	0.03	0.11	1,404
% Age ∈ (60,69)	0.13	0.04	0.12	1,404
% Age ∈ (70,79)	0.14	0.05	0.13	1,404
% Age ∈ (80,89)	0.11	0.06	0.10	1,404
% Age > 89	0.03	0.02	0.02	1,404
% Public Insurance	0.97	0.04	0.98	1,404
Hospital Characteristics:				
In-hospital Death Rate	3.32	1.81	2.95	1,404
28-day Death Rate	5.22	2.72	4.63	1,404
Inpatient Admissions	5,664	7,683	1,932	1,404
Average Length of Stay	3.52	2.13	3.00	1,404
Readmission Rate	3.47	1.39	3.27	1,404
Number of Physicians	77.54	119.65	20.00	1,404
Average Scores	72.00	6.07	72.34	1,404

Panel B: Test Scores					
Year	Number Tests Takers	Average score	% Approved (score ≥ 51)	Average score if score ≥ 51	# Tests ∈ [51 – 56)
2009	1,389	71.8	92	74.3	70
2010	1,535	65.1	80	72.1	88
2011	1,748	66.6	81	73.3	104
2013	2,003	56.1	66	67.5	129
2014	2,557	55.8	65	67.5	203
2015	3,641	54.7	60	66.5	323
2016	4,998	53.0	54	66.9	466
2017	6,014	52.1	55	64.9	662
2018	7,121	53.9	58	65.0	856

Notes: This table presents descriptive statistics for the data used in the primary analysis. Panel A summarizes patient and hospital characteristics for all public hospitals included in the analysis. Data are derived from individual-level inpatient records reported by [DEIS \(2019\)](#) and restricted administrative records on public hospital employees from [SIRH \(2019\)](#). Panel B provides statistics on EUNACOM scores from 2009 to 2018. For cases in which the exam was taken twice in the same year, all data are pooled. The dataset includes all records of test-takers and is from [ASOFAMECH \(2019\)](#).

Table A.2: Shock Summary Statistics

	Quantity Shock		Quality Shock	
	(1)	(2)	(3)	(4)
Mean	4.5	0.0	-0.7	0.0
Standard deviation	18.7	18.5	3.3	2.0
Interquartile range	1.8	2.6	6.1	2.0
Residualizing on period FE	No	Yes	No	Yes
Effective sample size ($1/HHI$ of s_{nt} weights)	29.35	29.35	29.35	29.35
Largest s_{nt} weight	0.06	0.06	0.06	0.06
No. of regions-period shocks	168	168	168	168
No. of regions	21	21	21	21

Notes: This table summarizes the distribution of migration shocks across regions of origin and periods. Quantity shocks are measured as the percentage change in the number of physicians from each region of training who clear the cutoff of the licensing exam. Similarly, quality shocks are measured as the change in the average quality of eligible test-takers from different regions of origin. All statistics are weighted by region-of-origin exposure shares. Columns (1) and (2) consider the quantity shocks, and Columns (3) and (4) the quality shocks. Columns (2) and (4) residualize migration shocks on period indicators. Following [Borusyak et al. \(2022\)](#), we report the effective sample size (the inverse re-normalized Herfindahl index of the s_{nt} weights) along with the largest s_{nt} .

Table A.3: Score Model: Time-Invariant and Trend Parameters

Panel A: Time-Invariant & Common Parameters

	Foreigners	Natives
$\hat{\pi}$ (weight, comp. 1)	0.87 (0.03)	0.95 (0.01)
$\hat{\mu}_{\theta,2}$ (comp. 2)	60.19 (1.09)	41.05 (0.76)
$\hat{\sigma}_{\theta,2}$ (comp. 2)	2.96 (1.33)	2.48 (0.90)
$\hat{\gamma}$ (common)	12.67 (0.68)	
$\hat{\rho}$ (common)	0.41 (0.06)	
Observations	7,273	9,725

Panel B: Structural Trend Parameters

	Foreigners	Natives
$\log \sigma_{\theta,1}$ (<i>comp. 1</i>), <i>piecewise</i>		
intercept (2013)	2.02 (0.18)	2.01 (0.17)
slope 2013-15	0.23 (0.07)	0.04 (0.10)
2015 deviation	—	-0.15 (0.09)
intercept (2016)	2.62 (0.03)	1.91 (0.11)
slope 2016-18	-0.02 (0.03)	-0.05 (0.09)
$\log \sigma_{\epsilon}$, <i>linear</i>		
intercept (2013)	1.88 (0.17)	1.75 (0.21)
slope (per year)	0.08 (0.04)	-0.02 (0.09)

Notes: This table reports the remaining parameters of the score model estimated by simulated method of moments. Panel A reports, separately by physician type, the weight on the first normal component, π , and the mean and standard deviation of the second component, $\mu_{\theta,2}$ and $\sigma_{\theta,2}$. It also reports the common parameters γ and ρ , which govern the initial gain in test-taking ability and its rate of decay over subsequent attempts. Panel B reports the parameters governing the type- and cohort-specific standard deviations of component-1 latent quality and score noise. We specify $\log \sigma_{\theta,1}$ as piecewise linear with a break in 2016 and a 2015 deviation for nationals, and $\log \sigma_{\epsilon}$ as linear. The estimation sample contains 7,273 foreign-trained and 9,725 nationally trained first-time test-takers from the 2013–2018 entry cohorts. Point estimates are bootstrap means, and standard errors in parentheses are standard deviations across 79 bootstrap iterations.

Table A.4: Impact of Physicians' Quantity and Quality: OLS

	Discharges	Mortality					Complications
	Ln Inpatient Admissions	Ln death rate In-Hospital	Ln death rate 28-Days	Ln Actual over Pred. death rate	Ln death rate Amenable	Ln death rate Non-amenable	Ln readmission rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ln Physicians ($\hat{\alpha}_L$)	0.05 (0.03)	0.03 (0.05)	-0.03 (0.05)	0.03 (0.05)	-0.09 (0.07)	0.12 (0.06)	0.07 (0.05)
Std. Physicians' Score ($\hat{\alpha}_\theta$)	-0.03 (0.02)	0.00 (0.03)	-0.03 (0.02)	-0.01 (0.02)	-0.03 (0.03)	0.00 (0.03)	-0.01 (0.02)
Observations	1,404	1,404	1,404	1,404	1,379	1,396	1,404
Mean Dep. Var. (levels)	5,664	3.32	5.22	0.75	10.47	2.11	3.47

Notes: This table presents the impact of the quantity and quality of physicians on public hospital performance. Estimates come from the OLS estimation of Equation (5). The set of conditions amenable to physician treatment are discussed in the main text. The mean dependent variable is presented in levels instead of logs. Robust standard errors in parentheses.

Table A.5: Impact of Physicians' Quantity and Quality: Recentered SSIV

	Discharges	Mortality					Complications
	Ln Inpatient Admissions	Ln death rate In-Hospital	Ln death rate 28-Days	Ln Actual over Pred. death rate	Ln death rate Amenable	Ln death rate Non-amenable	Ln readmission rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ln Physicians ($\hat{\alpha}_L$)	0.82 (0.27)	-0.81 (0.34)	-0.82 (0.31)	-0.74 (0.32)	-1.03 (0.40)	-0.16 (0.40)	-0.76 (0.33)
Std. Physicians' Score ($\hat{\alpha}_\theta$)	0.01 (0.08)	-0.17 (0.09)	-0.20 (0.09)	-0.21 (0.09)	-0.21 (0.12)	-0.09 (0.11)	-0.17 (0.09)
Observations	1,404	1,404	1,404	1,404	1,379	1,396	1,404
Mean Dep. Var. (levels)	5,664	3.32	5.22	0.75	10.47	2.11	3.47
F-stat (First-stage)	17.55	17.55	17.55	17.55	21.86	16.18	17.55
Anderson–Rubin (χ^2) p-value	0.00	0.01	0.00	0.00	0.00	0.65	0.01

Notes: This table presents the impact of the quantity and quality of physicians on public hospital performance. Estimates come from the re centered two-stage least squares estimation (Borusyak and Hull, 2023, 2025) of Equation (5). We compute the expected instruments under the assignment mechanism analytically: for each hospital and year, the expected quantity and quality instruments are the shift-share aggregates of the mean shocks within year-by-foreign-trained status cells, which we subtract from the realized instruments. The mean dependent variable is presented in levels instead of logs. Robust standard errors in parentheses.

Table A.6: Impact of Physicians' Quantity and Quality: Without Score Imputation

	Discharges	Mortality					Complications
	Ln Inpatient Admissions	Ln death rate In-Hospital	Ln death rate 28-Days	Ln Actual over Pred. death rate	Ln death rate Amenable	Ln death rate Non-amenable	Ln readmission rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ln Physicians ($\hat{\alpha}_L$)	1.00 (0.27)	-0.60 (0.18)	-0.64 (0.19)	-0.56 (0.16)	-0.85 (0.24)	-0.30 (0.24)	-0.46 (0.14)
Std. Physicians' Score ($\hat{\alpha}_\theta$)	-0.02 (0.04)	-0.09 (0.04)	-0.11 (0.05)	-0.11 (0.04)	-0.13 (0.04)	-0.05 (0.05)	-0.11 (0.03)
Observations	1,381	1,381	1,381	1,381	1,361	1,373	1,381
Mean Dep. Var. (levels)	5,745	3.33	5.22	0.76	10.50	2.11	3.46
F-stat (First-stage)	17.76	17.76	17.76	17.76	22.34	16.75	17.76
Anderson–Rubin (χ^2) p-value	0.00	0.01	0.00	0.00	0.00	0.48	0.01

Notes: This table presents the impact of physician quantity and quality on public hospital performance. Estimates come from two-stage least-squares estimation of Equation (5). The set of conditions amenable to physician treatment is discussed in the main text. In this case, we calculate average scores considering only physicians who took the exam at some point. The mean dependent variable is presented in levels instead of logs. We present exposure-robust standard errors clustered at the region-of-origin, i.e., the shock level, in parentheses (Adão et al., 2019; Borusyak et al., 2022).

Table A.7: Impact of Physicians' Quantity and Quality: Alternative Quality Index

	Discharges	Mortality					Complications
	Ln Inpatient Admissions	Ln death rate In-Hospital	Ln death rate 28-Days	Ln Actual over Pred. death rate	Ln death rate Amenable	Ln death rate Non-amenable	Ln readmission rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ln Physicians ($\hat{\alpha}_L$)	1.02 (0.24)	-0.50 (0.16)	-0.52 (0.16)	-0.45 (0.14)	-0.75 (0.21)	-0.25 (0.20)	-0.35 (0.14)
% Low Quality Physicians ($\hat{\alpha}_\theta$)	0.06 (0.17)	0.36 (0.16)	0.42 (0.17)	0.43 (0.14)	0.51 (0.18)	0.18 (0.17)	0.42 (0.13)
Observations	1,381	1,381	1,381	1,381	1,361	1,373	1,381
Mean Dep. Var. (levels)	5,745	3.33	5.22	0.76	10.50	2.11	3.46
F-stat (First-stage)	21.81	21.81	21.81	21.81	25.15	21.10	21.81
Anderson–Rubin (χ^2) p-value	0.00	0.01	0.00	0.00	0.00	0.48	0.01

Notes: This table presents the impact of physician quantity and quality on public hospital performance. Estimates come from two-stage least-squares estimation of Equation (5). The set of conditions amenable to physician treatment is discussed in the main text. In this case, we consider only physicians who took the exam at some point. The mean dependent variable is presented in levels instead of logs. We present exposure-robust standard errors clustered at the region-of-origin, i.e., the shock level, in parentheses (Adão et al., 2019; Borusyak et al., 2022).

Table A.8: Impact of Physicians' Quantity and Quality: Posterior Quality

	Discharges	Mortality					Complications
	Ln Inpatient Admissions	Ln death rate In-Hospital	Ln death rate 28-Days	Ln Actual over Pred. death rate	Ln death rate Amenable	Ln death rate Non-amenable	Ln readmission rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ln Physicians ($\hat{\alpha}_L$)	1.01 (0.32)	-0.69 (0.20)	-0.72 (0.23)	-0.67 (0.16)	-0.94 (0.30)	-0.37 (0.27)	-0.57 (0.15)
Posterior Quality ($\hat{\alpha}_\theta$)	-0.02 (0.07)	-0.15 (0.06)	-0.17 (0.07)	-0.17 (0.05)	-0.20 (0.07)	-0.08 (0.07)	-0.17 (0.05)
Observations	1,404	1,404	1,404	1,404	1,379	1,396	1,404
Mean Dep. Var. (levels)	5,664	3.32	5.22	0.75	10.47	2.11	3.47
F-stat (First-stage)	21.73	21.73	21.73	21.73	26.35	20.18	21.73
Anderson–Rubin (χ^2) p-value	0.00	0.01	0.00	0.00	0.00	0.47	0.01

Notes: This table presents the impact of physician quantity and quality on public hospital performance. Estimates come from two-stage least-squares estimation of Equation (5) using the measure of quality derived from the model of scores. The set of conditions amenable to physician treatment is discussed in the main text. The mean dependent variable is presented in levels instead of logs. We present exposure-robust standard errors clustered at the region-of-origin, i.e., the shock level, in parentheses (Adão et al., 2019; Borusyak et al., 2022).

Table A.9: Score Model: Targeted Moments, Data vs. Model

Moment	Nationals		Foreigners	
	Data	Model	Data	Model
Mean score, first attempt	67.73	67.35	41.48	41.30
S.D. score, first attempt	10.80	10.63	15.74	15.61
Mean score, retake	53.27	58.49	51.72	50.11
S.D. score, retake	9.53	9.24	14.02	14.07
Cov. (first, retake)	27.3	25.7	82.2	89.6
Pass rate ($s \geq 51$)	0.924	0.927	0.302	0.291
Pass rate ($s \geq 56$)	0.863	0.876	0.214	0.206
Share $46 \leq s < 51$	0.037	0.025	0.095	0.092
Share $51 \leq s < 56$	0.061	0.051	0.088	0.085
Share $56 \leq s < 61$	0.104	0.107	0.075	0.075
Mean score gain, retakers (pooled)	Data 15.45	Model 14.69		

Notes: This table compares the moments targeted in the simulated method of moments estimation with their model-implied counterparts, separately for nationally and foreign-trained physicians. First-attempt moments are averages across the 2013–2018 entry cohorts and are based on 9,725 nationally trained and 7,273 foreign-trained first-time test-takers. Moments involving a retake are computed from the pooled retaker samples for these cohorts. The final row pools both physician types.

Table A.10: Correlates of Elasticities

	Labor elasticity $\eta_{\underline{s}}^{L_{jt}}$				Quality Semi-elasticity $\eta_{\underline{s}}^{\bar{\theta}_{jt}}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
High (phys/pat) $_{j,t-1}$	0.048 (0.010)			0.044 (0.010)	-0.181 (0.041)			-0.179 (0.046)
High average score $_{j,t-1}$		0.025 (0.010)		0.022 (0.009)		-0.003 (0.043)		0.011 (0.042)
North $_j$			0.015 (0.010)	0.008 (0.010)			-0.046 (0.043)	-0.012 (0.045)
Mean Dep. Var.	-0.095	-0.095	-0.095	-0.095	0.353	0.353	0.353	0.353
Observations	1,056	1,056	1,056	1,056	1,056	1,056	1,056	1,056

Notes: The table shows OLS regression coefficients that relate the estimated quantity and quality elasticities to hospital observables. High (phys/pat) $_{j,t-1}$ is an indicator variable for hospitals above the median in the distribution of physicians per patient in the previous period. High average score $_{j,t-1}$ is an indicator variable for hospitals above the median in the distribution of average scores in the previous period. North $_j$ is an indicator variable for hospitals above the median in the distribution of latitudes. All specifications include year fixed effects.

Table A.11: The Impact of Accounting for General Equilibrium Effects in Matching Probabilities

Year	$\bar{\eta}_s^L$		$\bar{\eta}_s^\theta$		$\bar{\eta}_s^\theta / \bar{\eta}_s^L$		$\bar{\eta}_s^{\text{mortality}}$		$\bar{\eta}_s^{\text{access}}$	
	No GE	GE	No GE	GE	No GE	GE	No GE	GE	No GE	GE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2013	-0.0507	-0.0515	0.1370	0.1403	-2.7	-2.7	0.0102	0.0101	-0.0540	-0.0549
2014	-0.0502	-0.0572	0.1756	0.2050	-3.5	-3.6	0.0031	0.0026	-0.0544	-0.0620
2015	-0.0787	-0.0926	0.2741	0.3359	-3.5	-3.6	0.0050	0.0036	-0.0852	-0.1006
2016	-0.1094	-0.1415	0.4496	0.6148	-4.1	-4.3	-0.0050	-0.0122	-0.1199	-0.1558
2017	-0.1298	-0.1774	0.5012	0.7369	-3.9	-4.2	-0.0002	-0.0094	-0.1416	-0.1946
2018	-0.0680	-0.0822	0.2761	0.3540	-4.1	-4.3	-0.0025	-0.0066	-0.0744	-0.0904

Notes: The table quantifies the role of the general-equilibrium effects in matching probabilities. It shows the average elasticities for quantity (Cols. 1–2), quality (3–4), the ratio between the two (Cols. 5–6), mortality (7–8), and access (9–10), where “No GE” columns correspond to estimates abstracting from general-equilibrium effects, and “GE” are our main estimates that account for them.